

Fluoride vs. Biofilm

How Everyday Materials Fight Caries

Group 5

Lauren-Ashley Asbridge, Kimberly Clark, Caroline Dudish, Khalil Elhazmiri, Lillian Fantz, Chunghyun Lee, Kellen Norton, Hannah Gresham Putnam, and Rachel Dobbs Wineman

Faculty Mentor: Dr. Jauregui

Behavioral Objectives

1. **Describe** the biological structure of enamel and dentin and **explain** how cariogenic biofilms initiate demineralization through acid production, extracellular polysaccharides, and ecological shifts in microbial communities.
2. **Differentiate** the microbiological and biochemical mechanisms by which fluoride agents (NaF, SnF₂, CaF₂, SDF, varnishes, pastes, rinses, and GI release) disrupt biofilm metabolism, reduce acidogenicity, and facilitate remineralization.
3. **Analyze** the dose-response relationship of fluoride by identifying concentrations and forms required to inhibit biofilm activity, enhance mineral uptake, and maintain biological safety across patient populations.

Behavioral Objectives

4. **Compare and contrast** the clinical effectiveness of fluoride delivery systems (varnishes, rinses, pastes, SDF, GI restoratives) based on scientific evidence.
5. **Apply** a biologically informed clinical decision tree to **select** fluoride interventions tailored to patient-specific factors such as salivary function, biofilm activity, caries risk profile, and restorative requirements.

Behavioral Objective 1

Describe the biological structure of enamel and dentin and explain how cariogenic biofilms initiate demineralization through acid production, extracellular polysaccharides, and ecological shifts in microbial communities.

Kimberly Clark

Search Strategy



PICO: Not Applicable

Databases and Websites used

Pubmed

Mental Dental

ScienceDirect

Nature.com

Keywords

Dentin, Enamel, Dental Plaque, Demineralization

Pubmed search string

((("dentin" OR "Enamel")) AND (("Dental Plaque" OR "Biofilms" OR oral biofilm)) AND dental caries OR demineralization OR caries initiation)) AND ((acid production OR acidogenic bacteria OR "Lactic Acid") OR ("Polysaccharides, Bacterial" OR extracellular polysaccharides OR glucans) OR ("Microbiota" OR microbial shift OR ecological shift OR dysbiosis))

Search results

89 Results (3 used)

Highest level of evidence

Systematic Review

Enamel

- Composition by weight:
 - ~ 96% inorganic matrix (carbonate apatite)
 - ~ 3% water
 - ~ <1% organic matrix

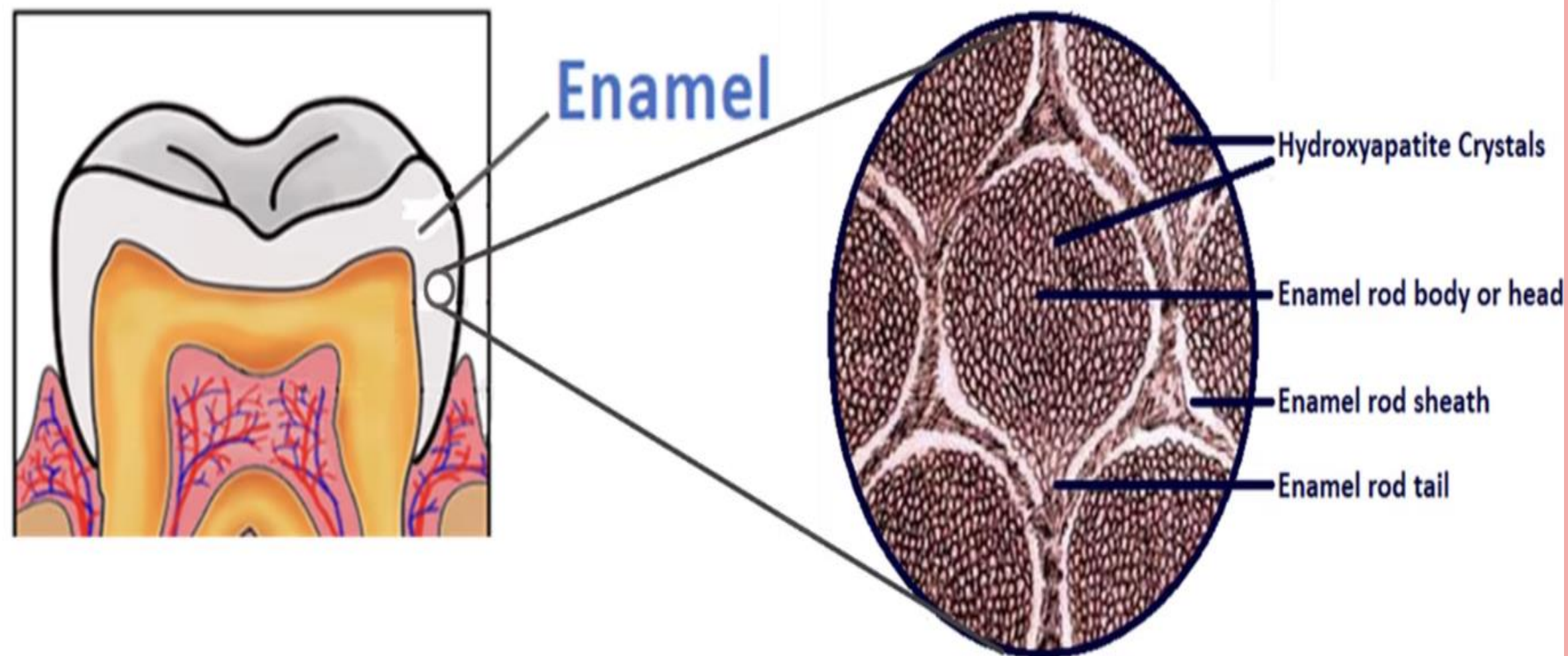
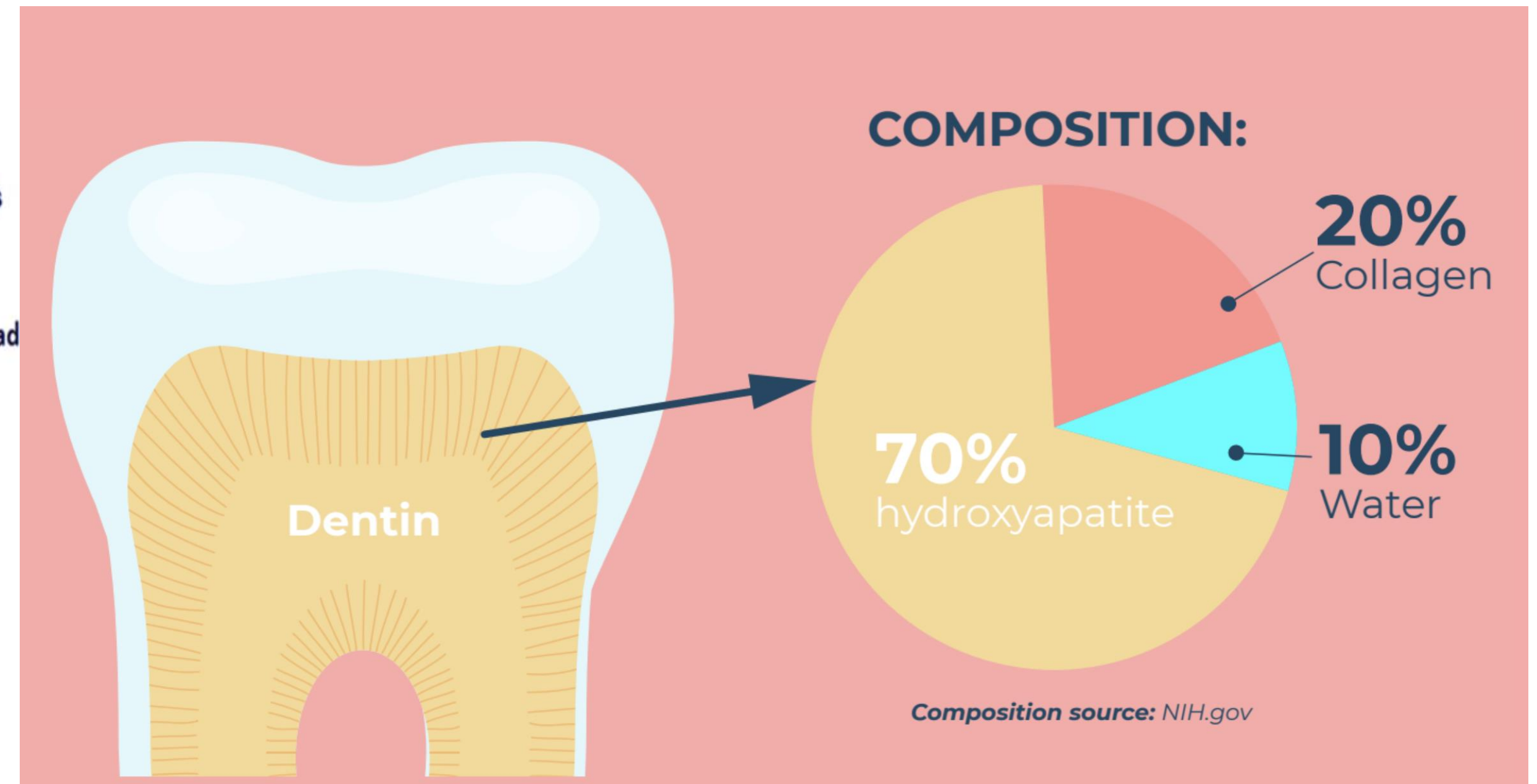


Figure 3. Enamel Structure and Composition

Source: Dentalcare.com

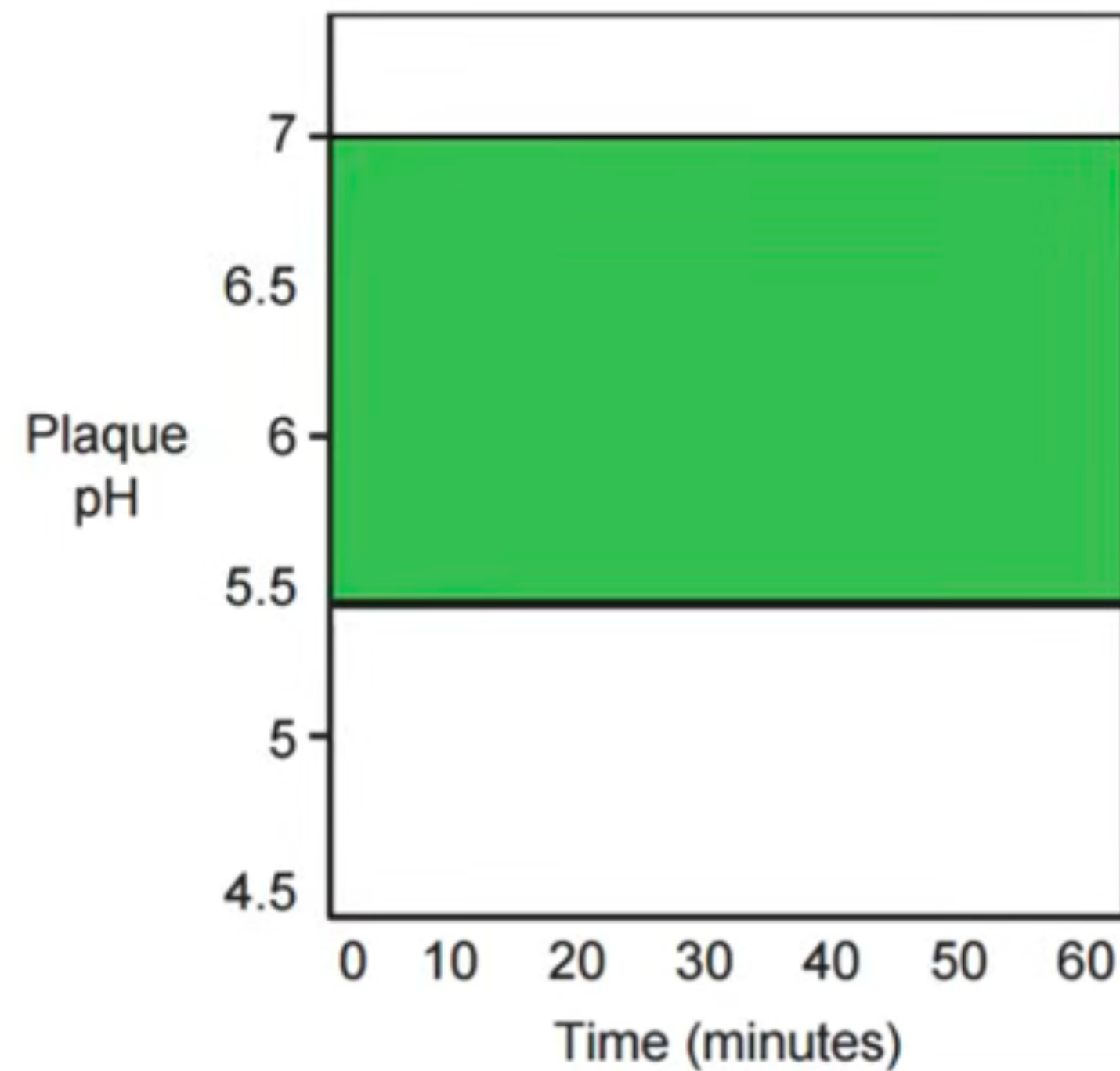
Dentin

- Composition by weight:
 - ~ 70% inorganic matrix (carbonate apatite)
 - ~ 10% water
 - ~ 20% organic matrix

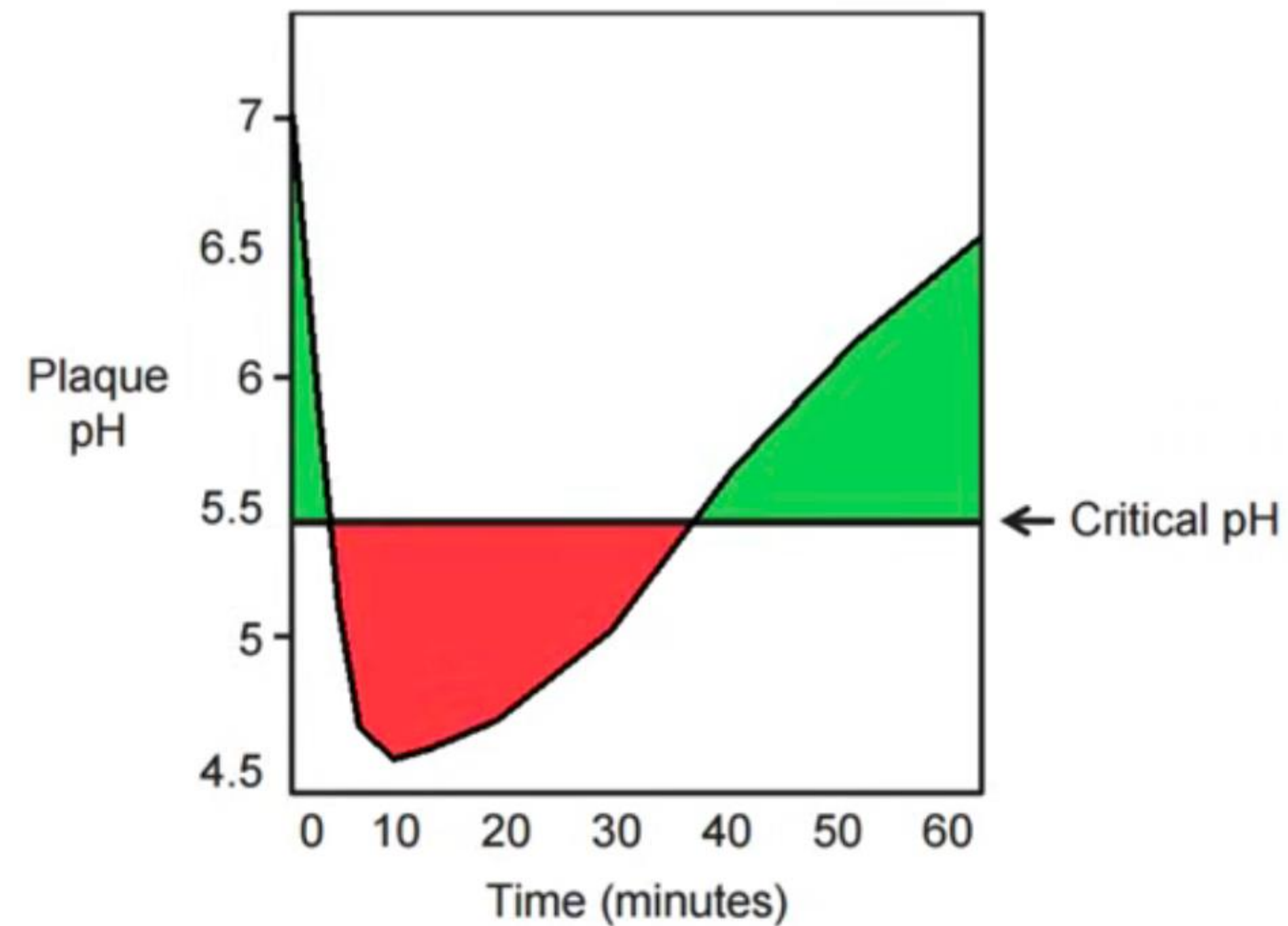




Stephan Curve

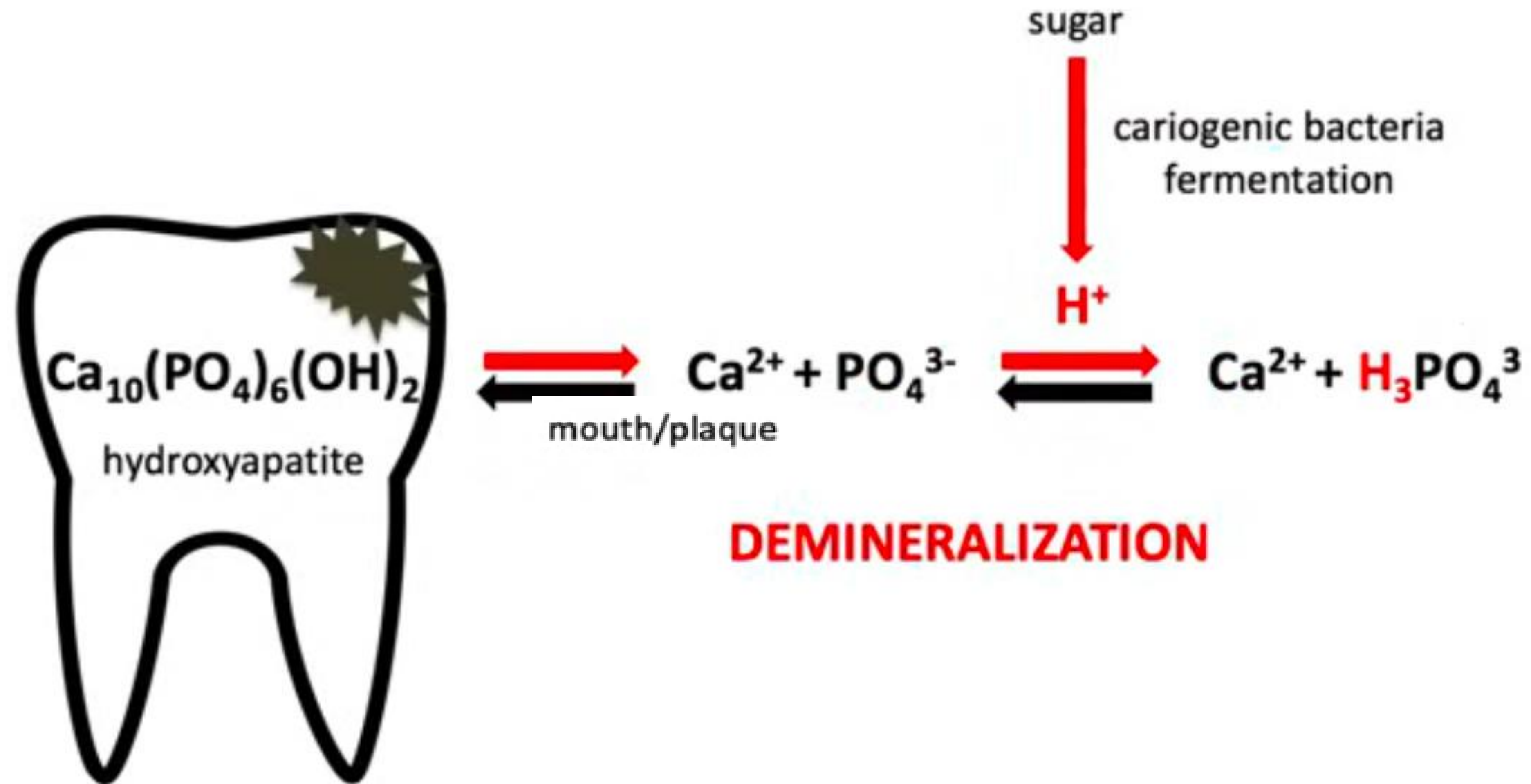


Stephan Curve (CHA)



- Critical pH:
 - 5.5 - Enamel
 - 6.2 - Dentin and Cementum

Biofilm, Acid Production, & Demineralization



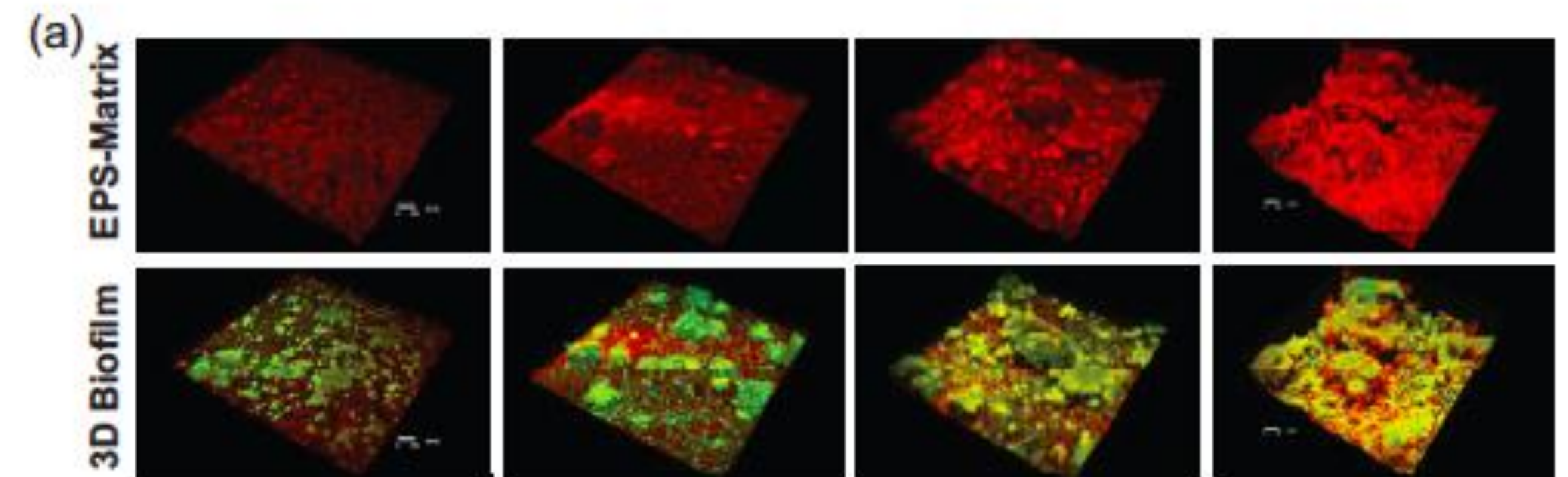
- Main Cariogenic bacteria: *S. mutans*, *S. sobrinus*, *Lactobacillus*



Extracellular Polysaccharides (EPS) & Demineralization



- Sucrose also used as a substrate for glucan synthesis via glucosyltransferases (Gtfs)
- Function
 - Adhesion and Cohesion
 - Matrix scaffold
 - Alters diffusion
 - Nutrient reserve
 - Promotes acid accumulation
- >3 genetically separate Gtfs
 - GtfB
 - GtfC
 - GtfD



Ecological Shift & Demineralization



- More acidic environment
- Acid-producing bacteria predominate vs. acid-tolerant bacteria
 - *S. mutans*, *Bifidobacterium* spp. , and Firmicutes
- Microbial imbalance leading to a cariogenic state of biofilm

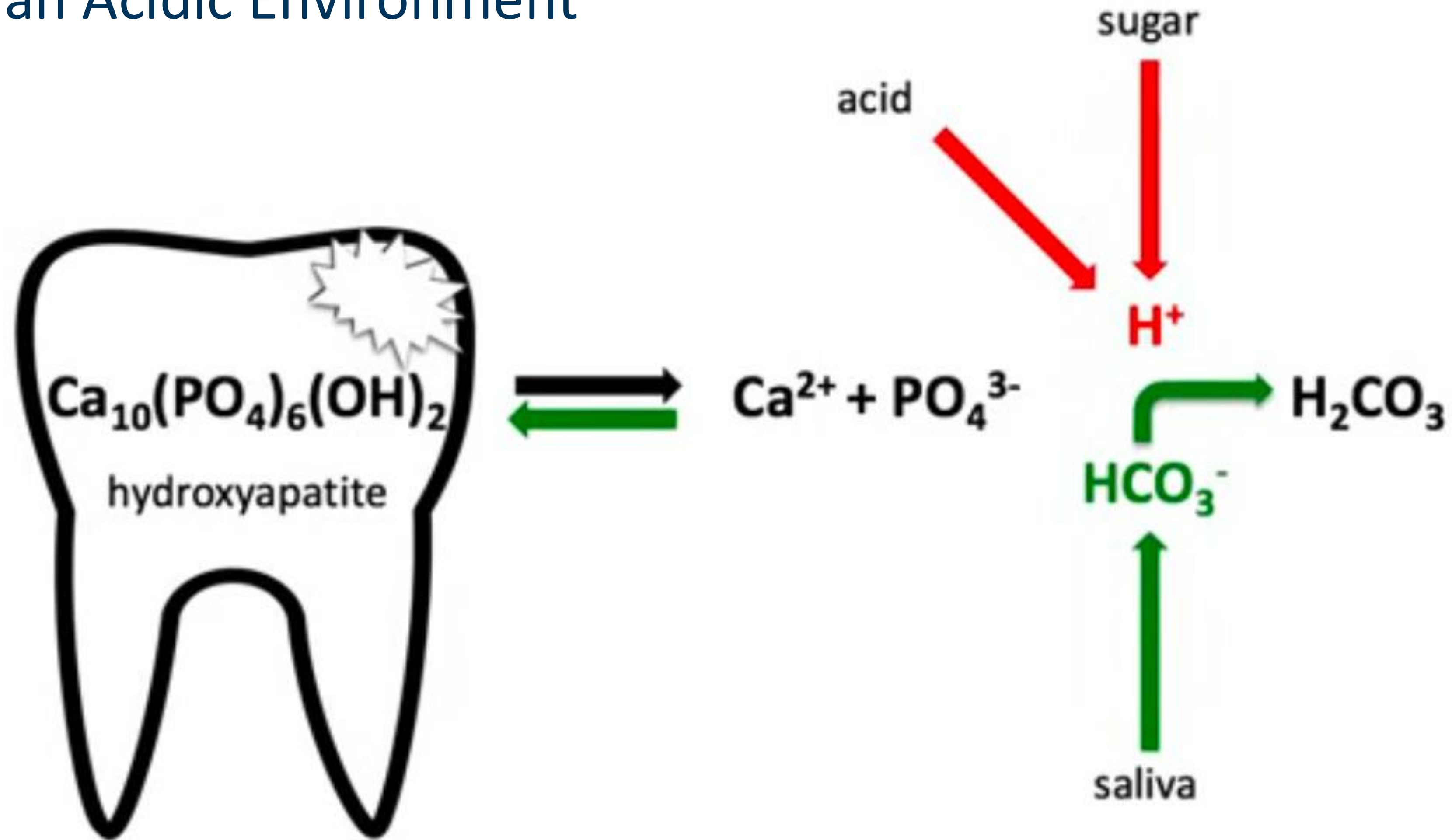
How does acid production & ecological shift relate to RSDM5511?



- Increase in enamel roughness
 - Increased friction and wear
 - Increased brittleness
 - Increases bacterial adhesion
 - Decreased resistance to decay
- Development of white spot lesions
 - Surface more porous → Scatters light differently → More opaque/ less translucent



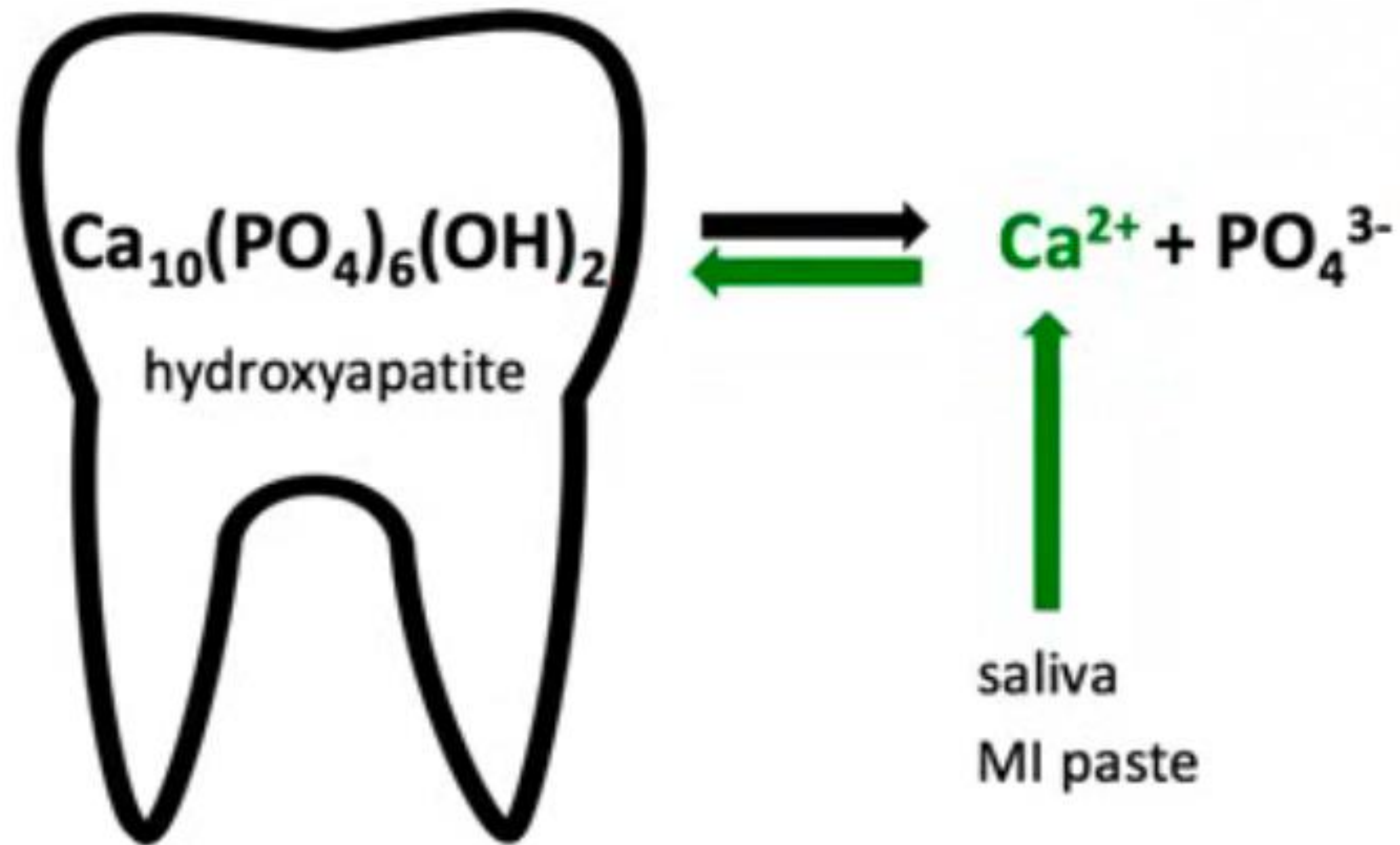
Combating an Acidic Environment



Remineralization Process



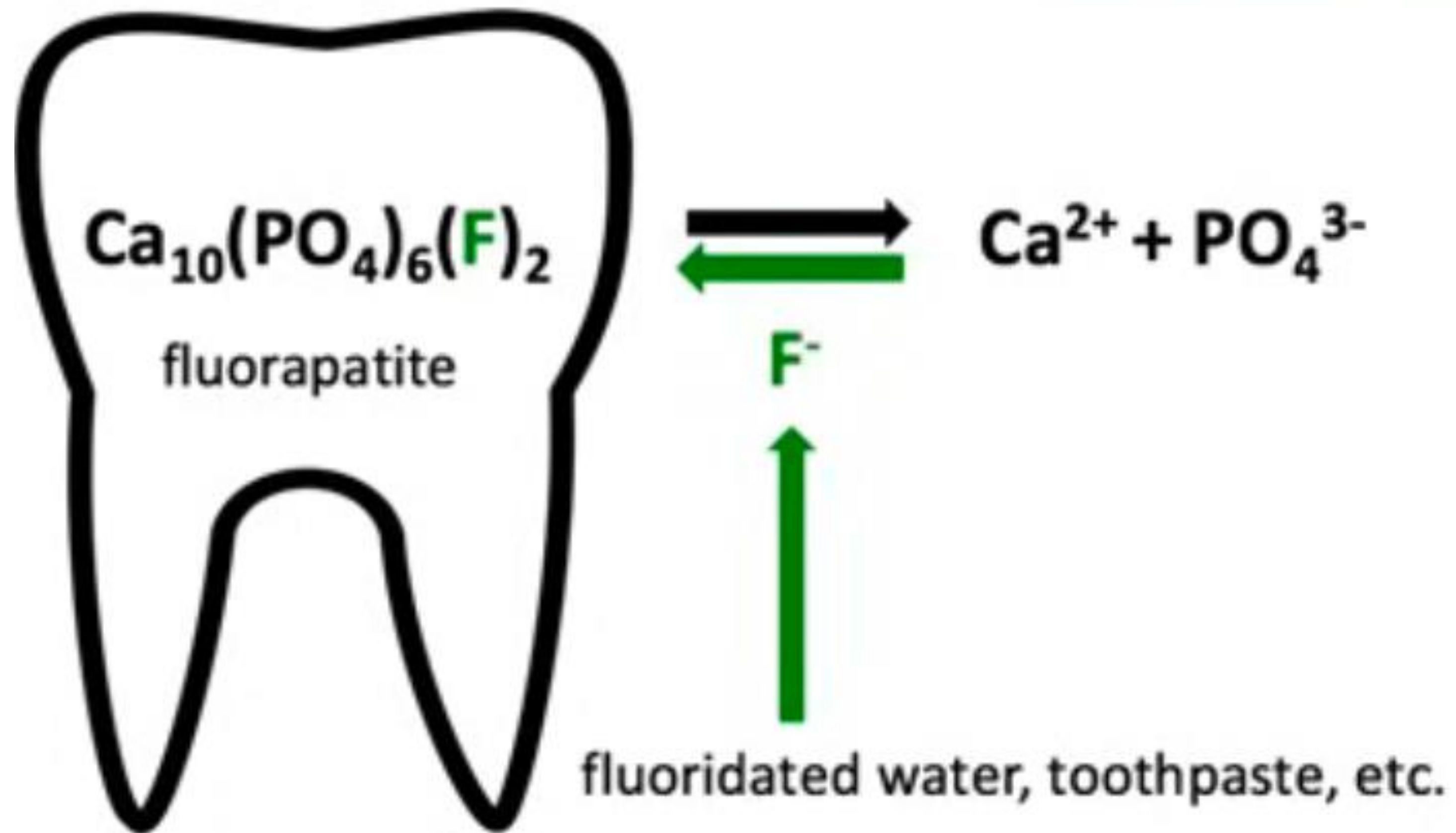
REMINERALIZATION



Ion Substitution



REMINERALIZATION



Behavioral Objective 2

Differentiate the microbiological and biochemical mechanisms by which fluoride agents (NaF, SnF₂, CaF₂, SDF, varnishes, pastes, rinses, and GI release) disrupt biofilm metabolism, reduce acidogenicity, and facilitate remineralization.

Lauren-Ashley Asbridge & Khalil Elhazmiri

Search Strategy



P: Teeth exposed to cariogenic biofilms

I: The application of fluoride agents

C: Compared with no fluoride or non-fluoride treatments

O: Caries Incidence

Databases used

Pubmed

Google Scholar

Trip Database

Galileo

Keywords

NaF, SnF₂, CaF₂, SDF, varnishes, pastes, rinses, or GI release, no fluoride, biofilm, metabolism, acidogenicity

Search results

242 found, 21 used, 2 discussed

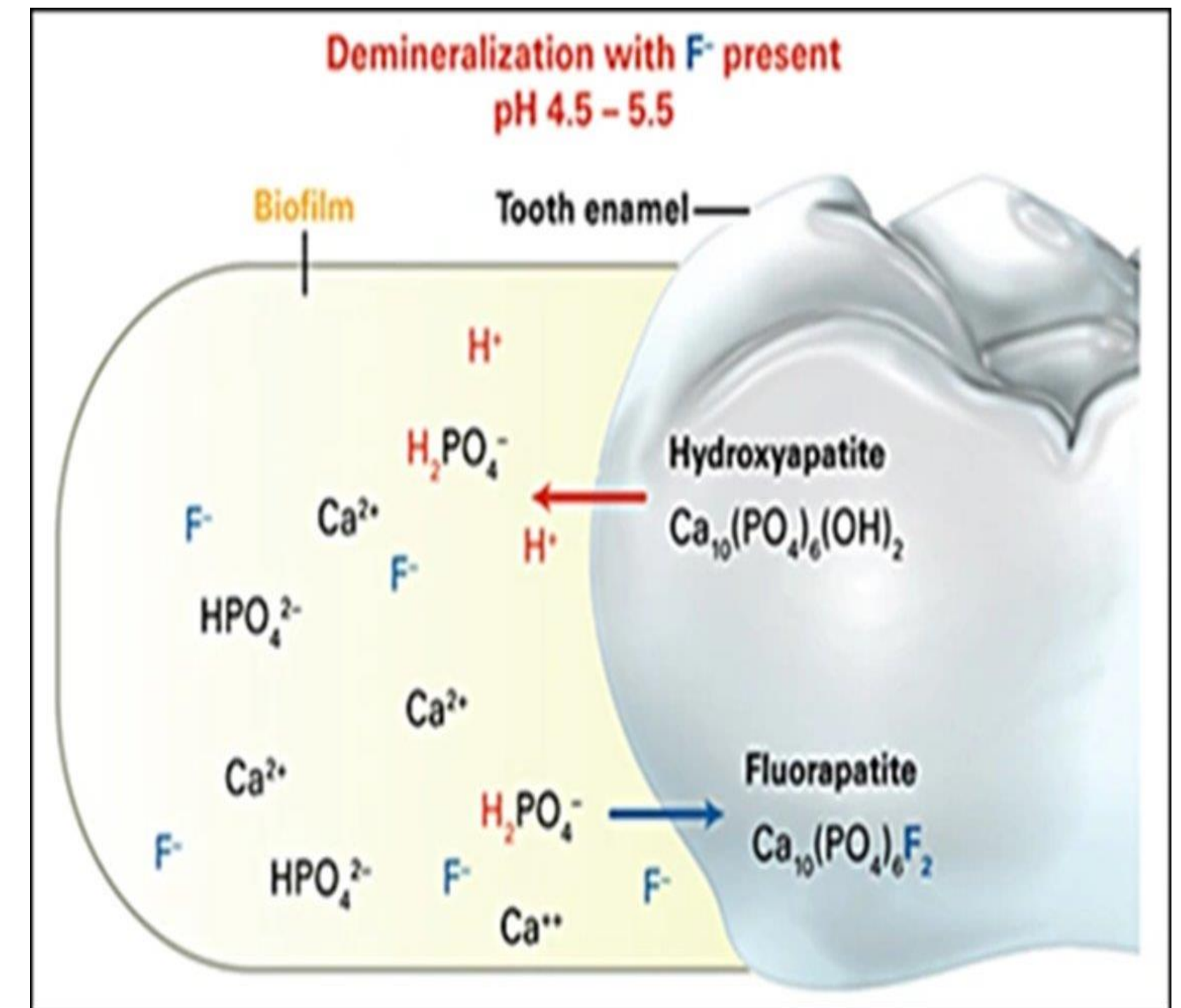
Highest level of evidence

Systematic reviews



Sodium Fluoride

- Inhibits bacterial glycolysis (enolase) → ↓ acid production
- Disrupts pH regulation in *S. mutans*
- Forms CaF_2 -like reservoirs releasing F^- during acid attack
- Promotes fluorapatite formation → stronger enamel





Stannous Fluoride (SnF_2)

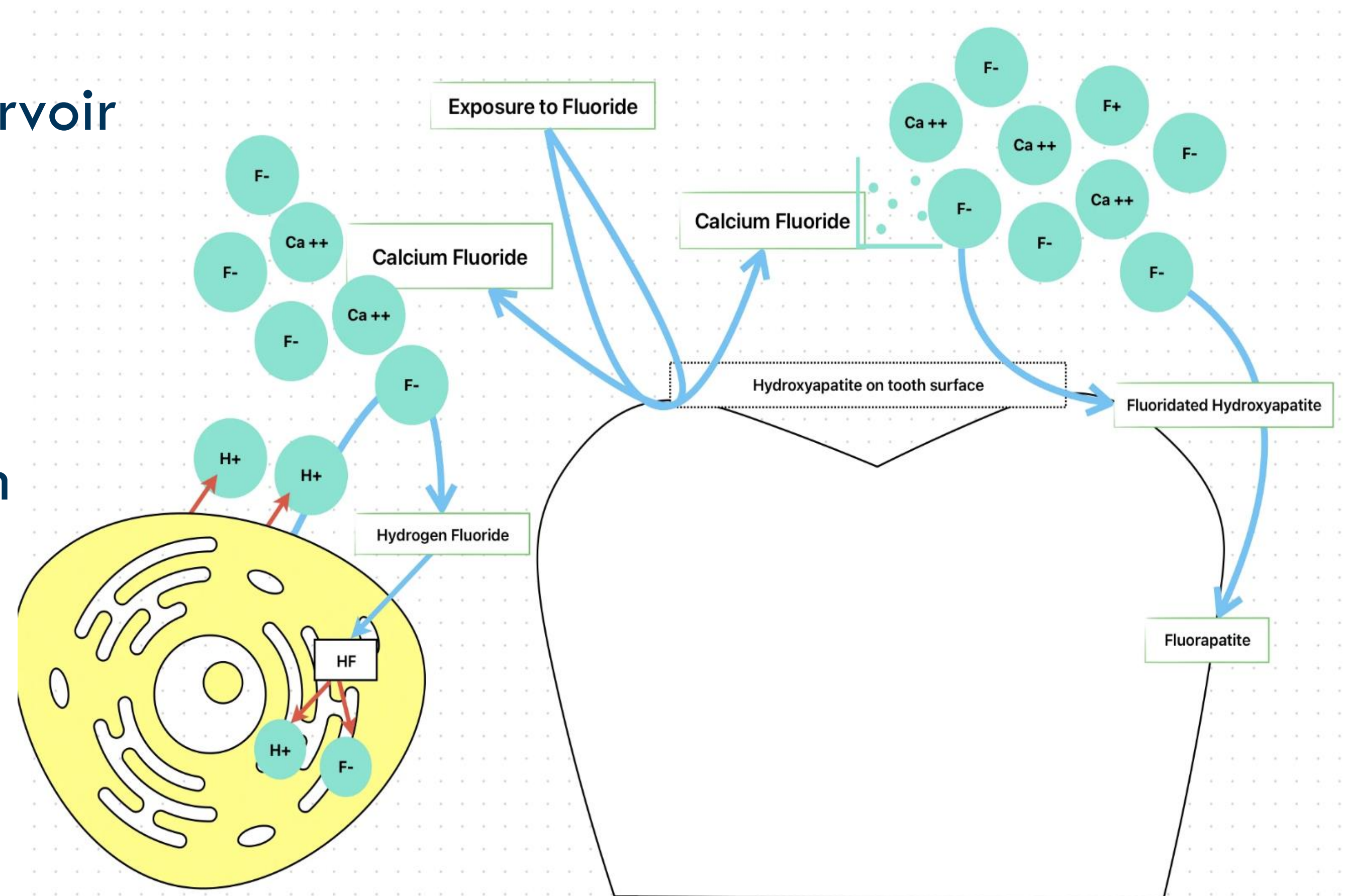
- Tin (Sn^{2+}) disrupts bacterial membranes & enzymes
- Inhibits EPS formation → weaker biofilm
- Provides fluoride for remineralization
- Reduces MMPs → protects dentin collagen





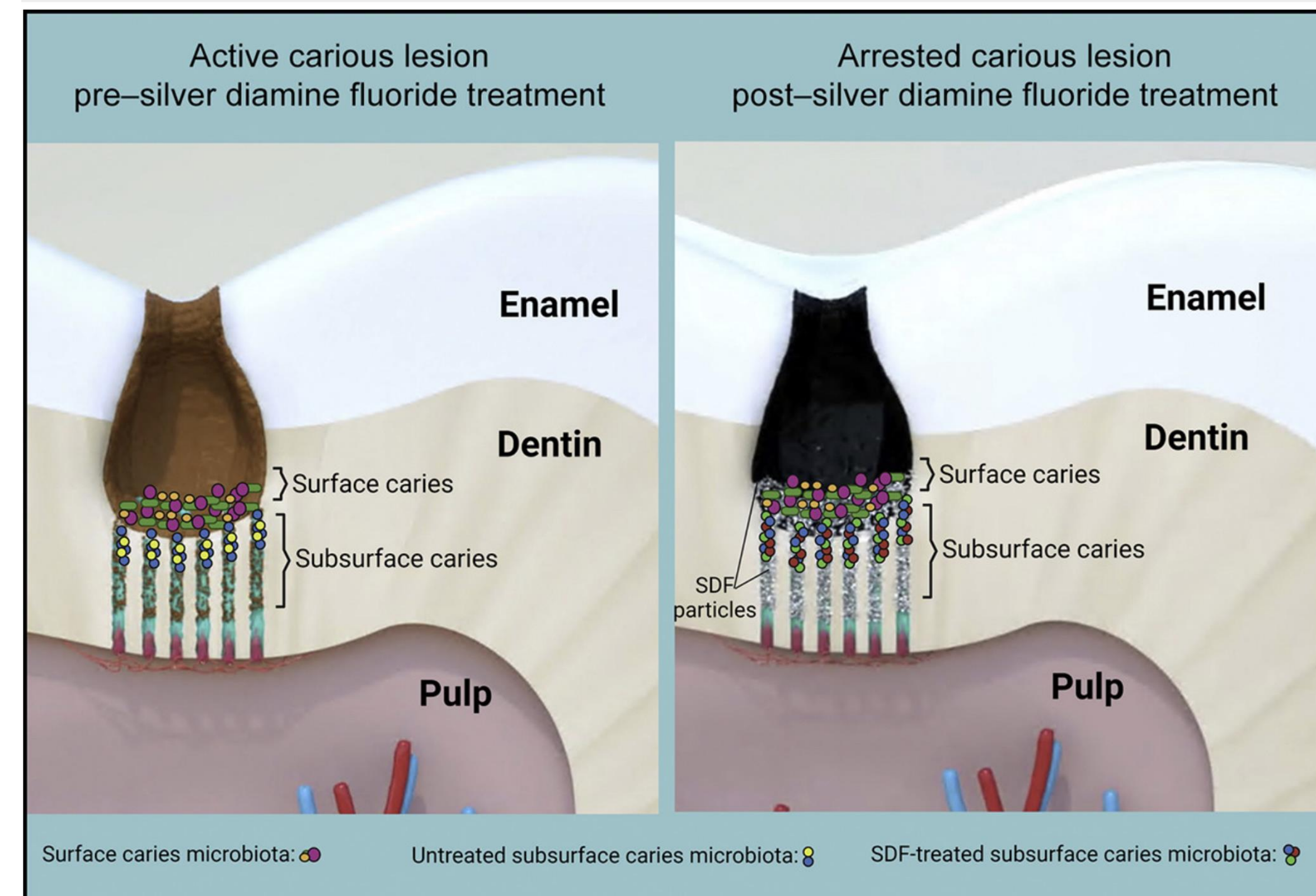
Calcium Fluoride (CaF_2)

- Acts as slow-release fluoride reservoir
- Releases F^- during acid attacks
- Encourages fluorapatite formation



Silver Diamine Fluoride (SDF) "The Silver Bullet"

- Silver kills bacteria & denatures proteins
- Fluoride promotes remineralization
- Hardens carious dentin (silver phosphate complexes)
- Inhibits collagen breakdown





Fluoride Varnishes

- **Concentration:** 5% Sodium Fluoride (NaF) = 22,600 ppm fluoride
- **Mechanism:** Sustained fluoride, builds protective CaF_2 layer on enamel, Reduces bacterial metabolism and acid production
- **Formulations:**
 - Rosin-based varnish (traditional): adheres well, releases fluoride gradually
 - Polyacrylic acid-based varnish: improved adhesion and fluoride release in moist conditions

Fluoride Pastes & Dentifrices



Primary Action: Inhibit bacterial metabolism, Promotes fluorapatite formation

- **Common Fluoride Types:**
- Sodium Fluoride (NaF) – stable, rapid fluoride release
- Stannous Fluoride (SnF_2) – antibacterial, strengthens enamel
- Sodium Monofluorophosphate (MFP) – slower fluoride release, gentle for children
- **Formulation Types:**
- Whitening (peroxide or abrasives)
- Desensitizing (potassium nitrate, stannous fluoride)
- Antibacterial (stannous fluoride, triclosan – limited use)
- Natural/Herbal (fluoride or fluoride-free)
- Children's (lower fluoride concentration, mild flavor)



Fluoride Mouthrinses

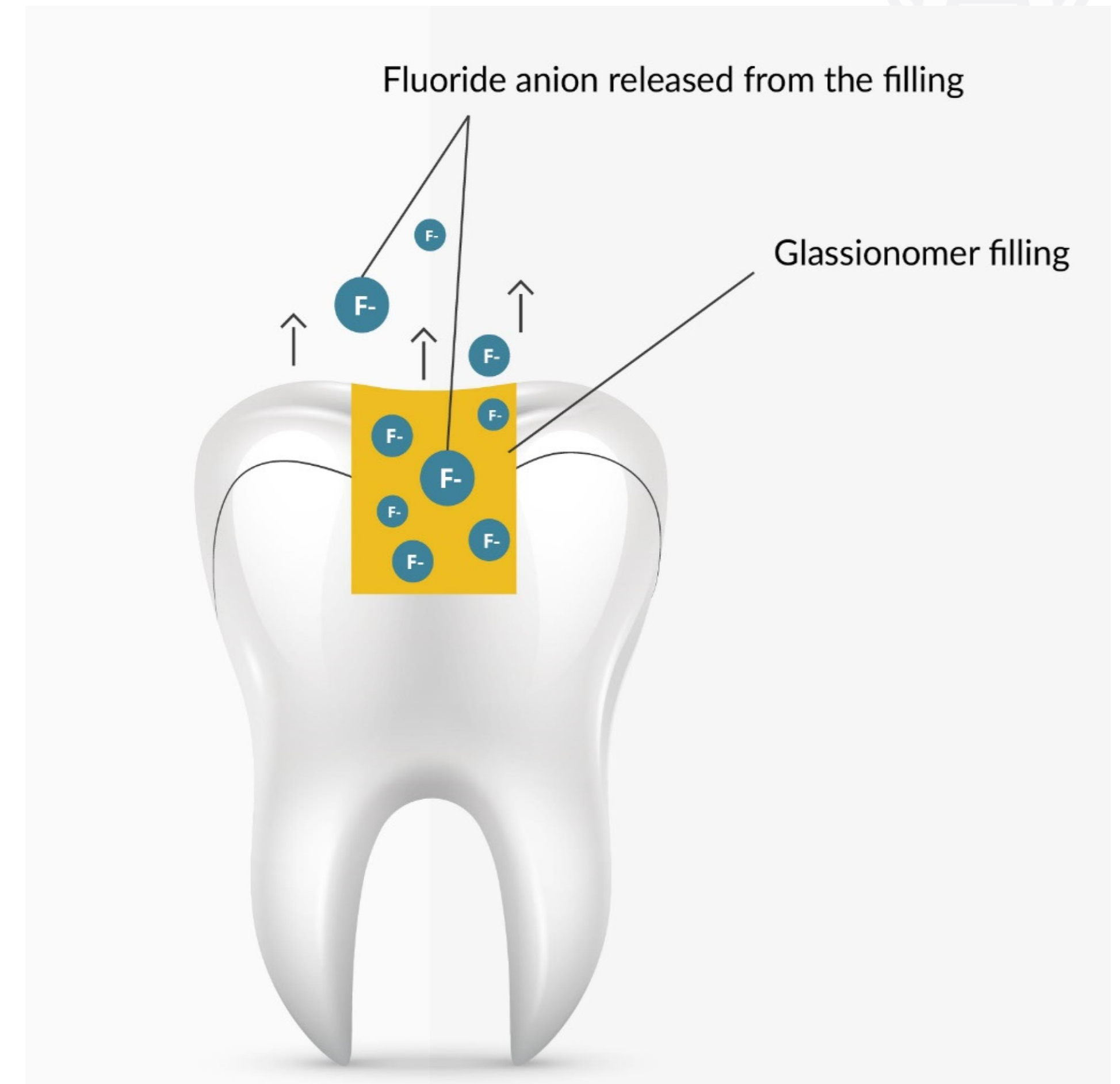
- **Primary Action:** Inhibit bacterial metabolism, Promotes fluorapatite formation
- Can provide a quick and on the go source of fluoride
- CaF_2 -like deposits and RMGI recharge
- Strengthen enamel between brushings





Glass Ionomer (GI) Fluoride Release

- Reaction between fluoroaluminosilicate glass and polyacrylic acid
- Continuous low-level fluoride release
- Fluoride “recharges” from toothpaste/rinses
- Remineralizes surrounding enamel/dentin





SUMMARY! How Fluoride Agents Disrupt Biofilm and Promote Remineralization

Agent	Antimicrobial/Biofilm Effect	Acid Reduction Mechanism	Remineralization Mechanism
NaF	Inhibits enolase, reduces metabolism	Lowers acidogenicity	Fluorapatite formation, CaF ₂ storage
SnF₂	Tin ions disrupt biofilm & inhibit EPS	Inhibits acidogenic enzymes	Forms stannous deposits & fluorapatite
CaF₂	Fluoride release inhibits bacteria	Buffers acid by acting at low pH	Fluoride reservoir → remineralization
SDF	Silver kills bacteria	Dual silver & fluoride suppression	Hardens dentin, forms stable complexes
Varnish	Sustained fluoride inhibits biofilm	Long-term acid suppression	Deep remineralization, CaF ₂ layer
Paste	Short-term antibacterial	Inhibits glycolysis post-brush	Rapid fluoride uptake, enamel repair
Rinse	Metabolism inhibition in plaque fluid	Reduces acid production	Fluorapatite & CaF ₂ formation
GI Release	Continuous fluoride release suppresses biofilm	Persistent acidogenicity reduction	Sustained remineralization via fluoride recharge



Effectiveness of SDF vs NaF varnish/MI for arresting early childhood caries



National Library of Medicine
National Center for Biotechnology Information

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Advanced User Guide



Randomized Controlled Trial > BMC Oral Health. 2023 Oct 4;23(1):710.
doi: 10.1186/s12903-023-03456-3.

Effectiveness of silver diamine fluoride versus sodium fluoride varnish combined with mother's motivational interviewing for arresting early childhood caries: a randomized clinical trial

Randa Yassin¹, Hala Amer², Maha El Tantawi²

Table 2.

Caries arrest rates of active caries lesions in the study groups after 6 months in bivariate analysis, overall and by baseline lesion severity

	SDF group n/N (%)	NaF/ MI group n/N (%)	P value
All lesions	311/488 (63.7)	268/461 (58.1)	0.08
Moderate (ICDAS 3 or 4) lesions	97/133 (72.9)	133/191 (69.6)	0.52
Advanced (ICDAS 5 or 6) lesions	214/355 (60.3)	135/270 (50.0)	0.01* 

(Yassin R et. al., 2023)

SnF2-containing vs NaF-containing dentifrices in Reducing Erosive Wear



Randomized Controlled Trial > Caries Res. 2011;45(6):518-23. doi: 10.1159/000331391.
Epub 2011 Oct 5.

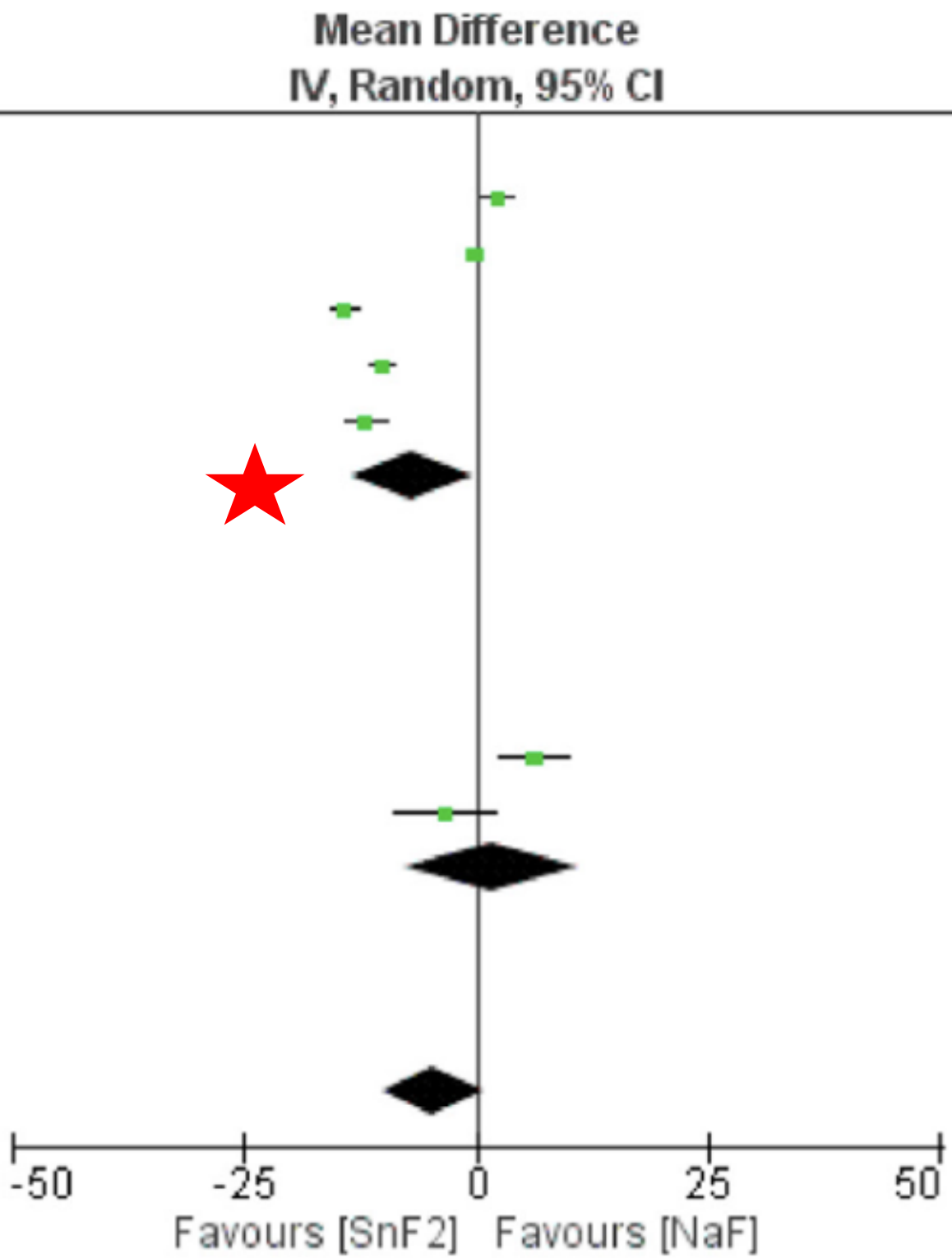
Reduction of erosive wear in situ by stannous fluoride-containing toothpaste

M C D N J M Huysmans¹, D H J Jager, J L Ruben, D E M F Unk, C P A H Klijn, A M Vieira

Affiliations + expand

PMID: 21985895 DOI: 10.1159/000331391

Study or Subgroup	SnF2			NaF			Weight	Mean Difference IV, Random, 95% CI
	Mean	SD	Total	Mean	SD	Total		
1.1.1 <i>in vitro</i> studies								
Assunção et al., 2019	17.1	2.2	17	14.8	2.8	17	14.6%	2.30 [0.61, 3.99]
Comar et al., 2012	2.8	1.1	12	2.9	0.7	12	14.8%	-0.10 [-0.84, 0.64]
Ganss et al., 2011	6.3	2.1	18	20.5	5.2	90	14.7%	-14.20 [-15.65, -12.75]
Ganss et al., 2012	4.6	1.9	15	14.8	2.9	30	14.7%	-10.20 [-11.61, -8.79]
Schlueter et al., 2016	5.4	7.2	160	17.3	12.2	160	14.5%	-11.90 [-14.10, -9.70]
Subtotal (95% CI)			222			309	73.1%	-6.80 [-13.42, -0.19]
Heterogeneity: Tau ² = 56.25; Chi ² = 474.79, df = 4 (P < 0.00001); I ² = 99%								
Test for overall effect: Z = 2.02 (P = 0.04)								
1.1.2 <i>in situ</i> studies								
Huysmans et al., 2011	8.8	7.9	24	2.5	4.9	24	13.9%	6.30 [2.58, 10.02]
Zhao et al., 2017	9.1	13.8	48	12.5	13.8	48	13.0%	-3.40 [-8.92, 2.12]
Subtotal (95% CI)			72			72	26.9%	1.67 [-7.82, 11.17]
Heterogeneity: Tau ² = 41.28; Chi ² = 8.16, df = 1 (P = 0.004); I ² = 88%								
Test for overall effect: Z = 0.35 (P = 0.73)								
Total (95% CI)			294			381	100.0%	-4.54 [-10.18, 1.10]
Heterogeneity: Tau ² = 55.93; Chi ² = 504.07, df = 6 (P < 0.00001); I ² = 99%								
Test for overall effect: Z = 1.58 (P = 0.11)								
Test for subgroup differences: Chi ² = 2.06, df = 1 (P = 0.15), I ² = 51.5%								



(Braz. oral. res. 35, 2021)

Behavioral Objective 3

Analyze the dose-response relationship of fluoride by identifying concentrations and forms required to inhibit biofilm activity, enhance mineral uptake, and maintain biological safety across patient populations.

Brooks Norton & Lillian Fantz

Search Strategy



P: Dental Caries

I: Fluoride

C: No or Limited Fluoride

O: *

Databases used

Pubmed

Google Scholar

Trip Database

Galileo

Keywords

Fluoride, Caries Incidence, Concentration, Fluorosis, Biofilm, Remineralization

Pubmed search string

((Fluoride Concentration) AND (Caries Incidence))

Search results

15 systematic reviews found

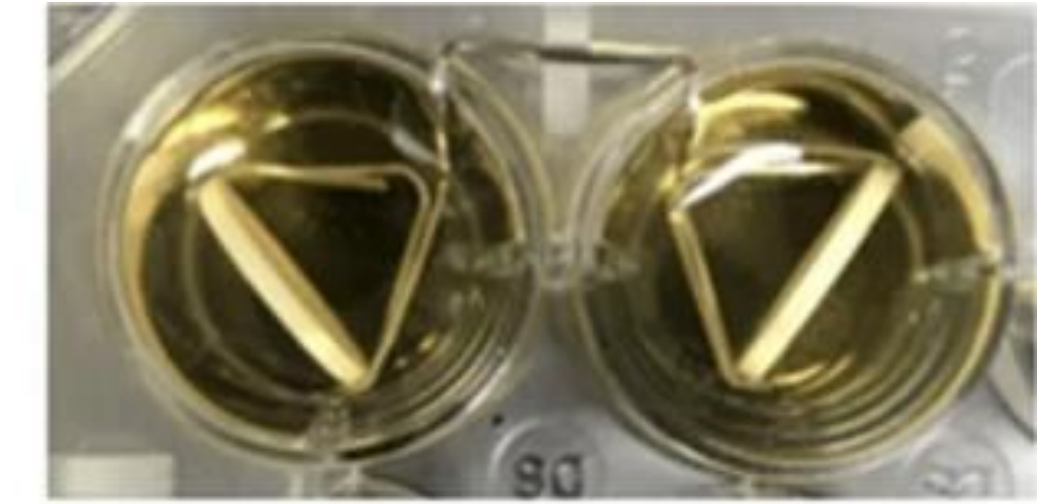
Highest level of evidence

Systematic reviews



Fluoride Concentrations and Form Summary

Delivery System	Typical ppm F	Key Effect	Safety Notes / Comments
Toothpaste	1,000–1,500 ppm (OTC); 5,000 ppm (Rx)	Daily baseline remineralization Bactericidal	Supervise < 6 yrs
Varnish	22,600 ppm (NaF)	Forms CaF ₂ reservoir	2–4× per year
SDF	44,800 ppm F (+ Ag ⁺)**	Arrests dentin caries	Staining + consent required (Alnasser et al., 2024)
GIC / RMGI	50–200 ppm initial burst → low-level release	Sustained fluoride protection (“smart release”)	
Rinses	225–900 ppm F	Maintenance / biofilm control	Avoid unsupervised use < 6 yrs

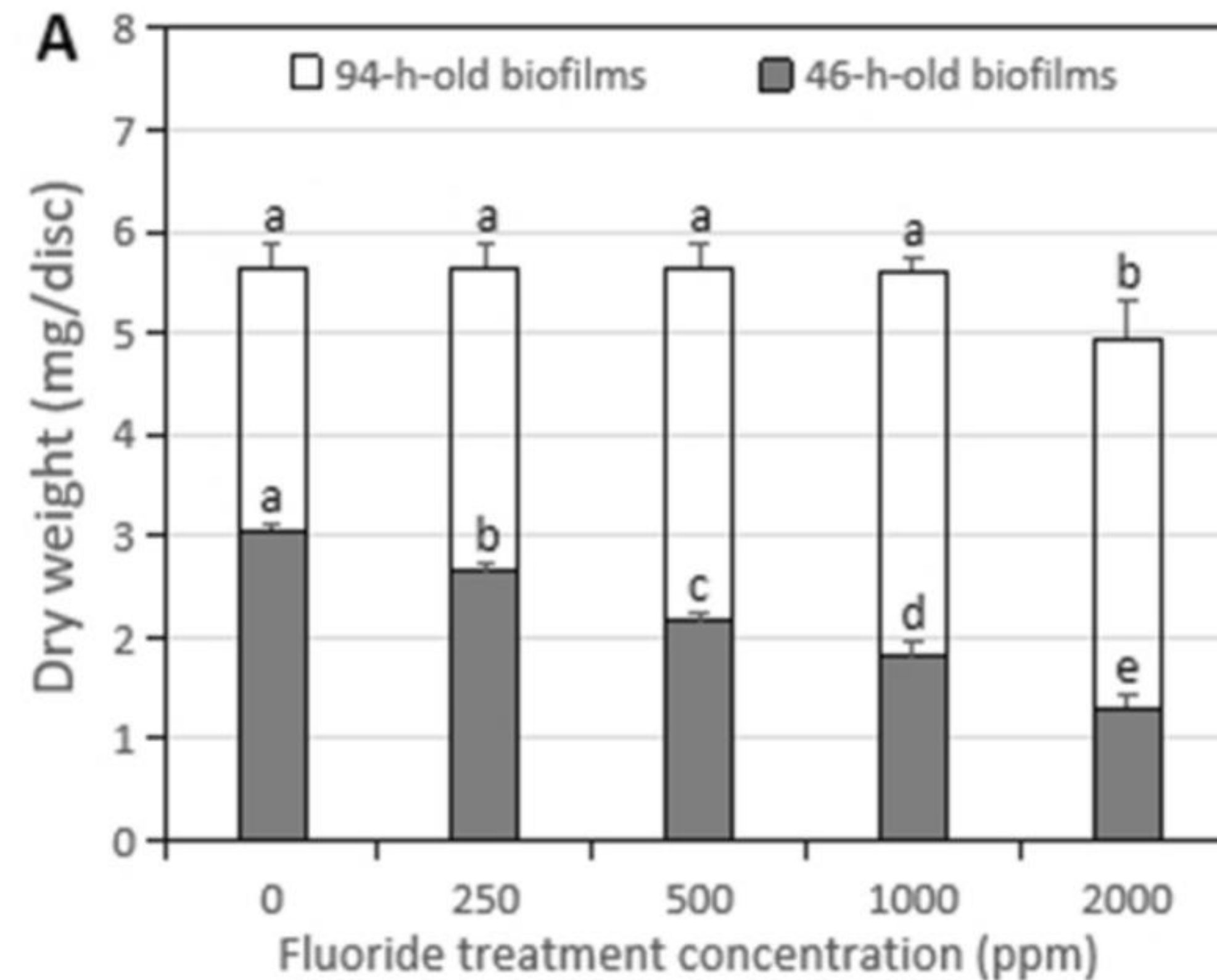


Fluoride Concentration and Biofilm Disruption: Nature 2021

- **Investigated antibiofilm activity of sodium fluoride during the early and mature stages of *S. mutans* biofilm formation.**
- **Early Stage:** All concentrations inhibited growth of *S. mutans* biofilms.
 - The antibiofilm and anti-EPS formation activities of the brief fluoride treatments increased with a concentration-dependent pattern.
- **Late Stage:** Only the 2000 ppm inhibited biofilm accumulation, activity, and intracellular/extracellular polysaccharide content.
- Fluoride treatment in the early stage of cariogenic biofilm formation may be an effective approach to controlling cariogenic biofilm development and preventing dental caries

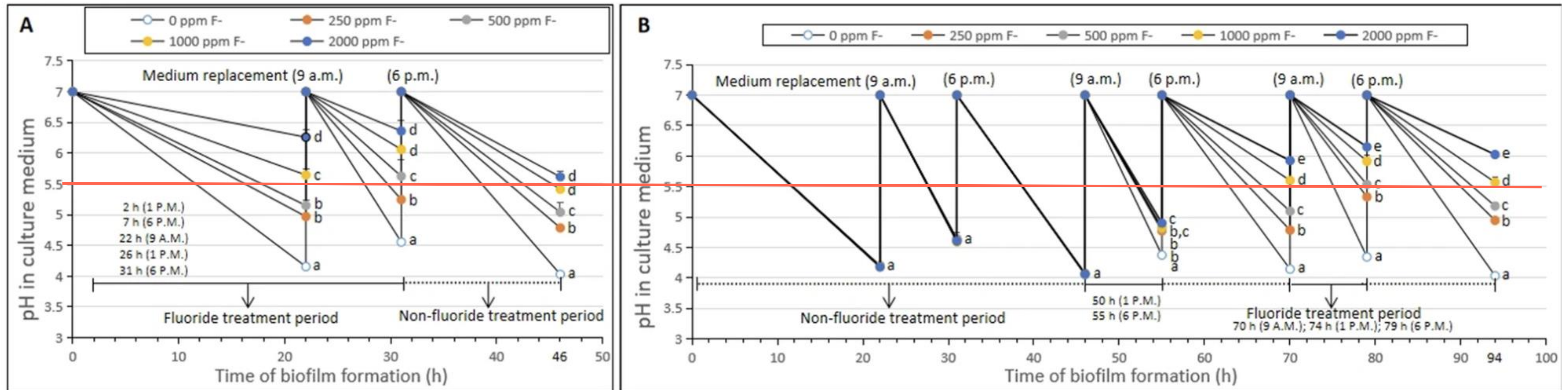


Fluoride Concentration and Biofilm Disruption: Weight





Fluoride Concentration and Biofilm Disruption: Acid Production

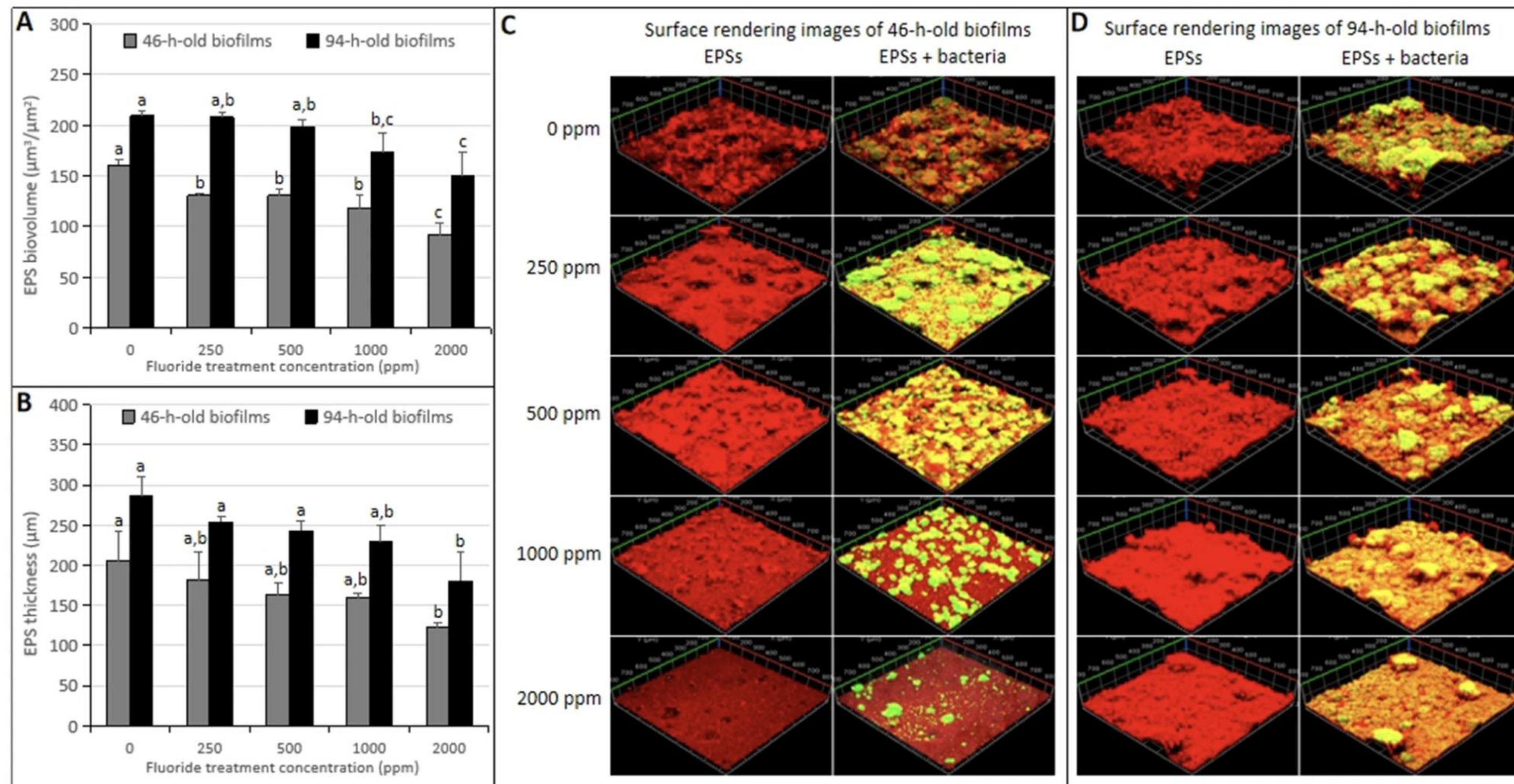


— Critical pH of enamel: 5.5





Fluoride Concentration and Biofilm Disruption: EPS





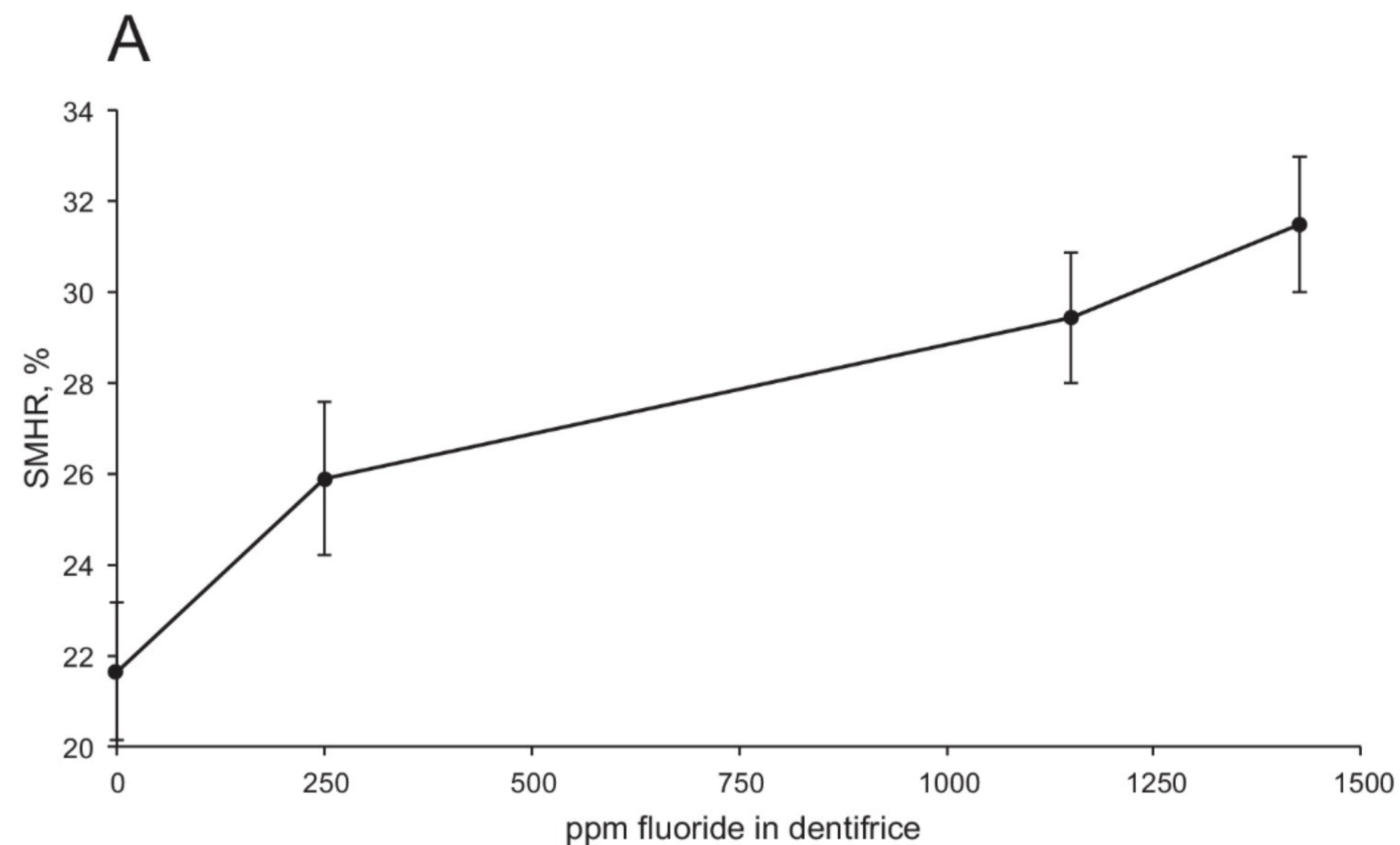
Fluoride Concentration and Mineralization Summary

Evaluate the dose–response effect of fluoride dentifrice on the remineralization of eroded enamel.

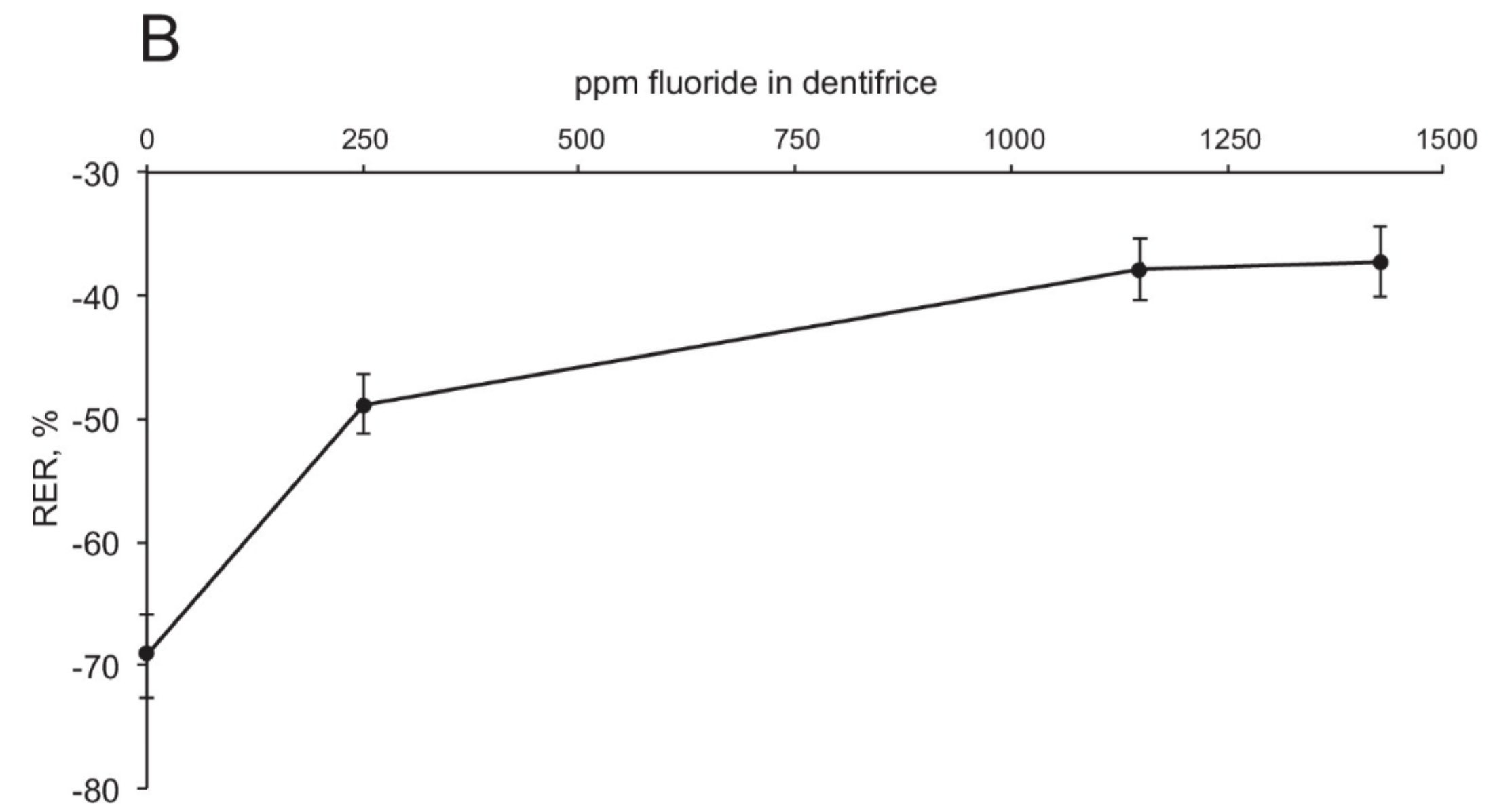
- Tested 0, 250, 500, 1 100, and 1 450 ppm Fluoride Concentrations
- Linear dose–response relationship between SMHR and dentifrice fluoride concentration
- Used analytical variables SMHR, RER, EFU, and CER to measure remineralization
 - **SMHR:** Surface Microhardness Recovery
 - **RER:** Relative Erosion Resistance
 - **EFU:** Enamel Fluoride Uptake
 - **CER:** Comparative Erosion Resistance



Fluoride Concentration and Mineralization



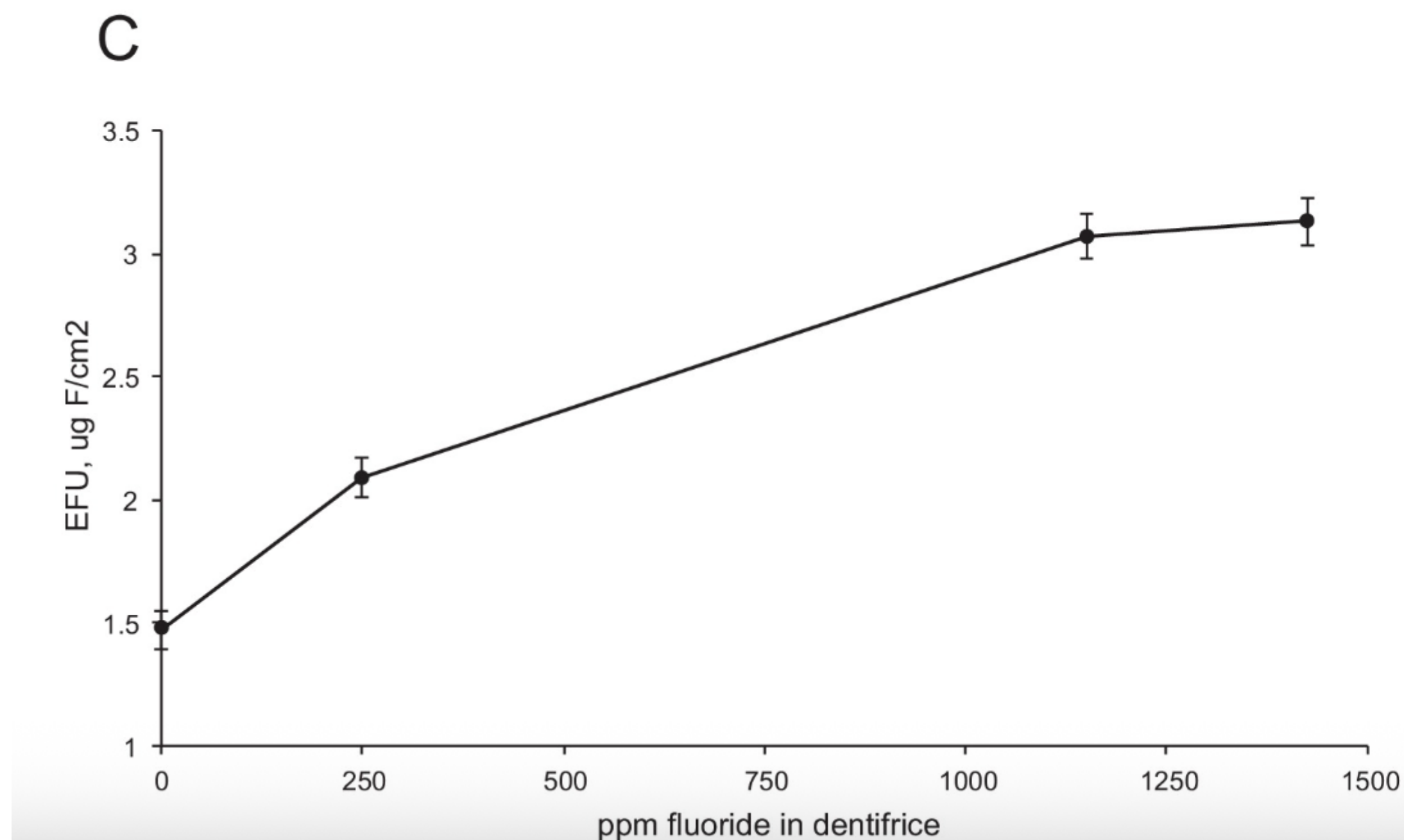
Dose-response increase: Enamel surface became harder and more remineralized with higher fluoride levels



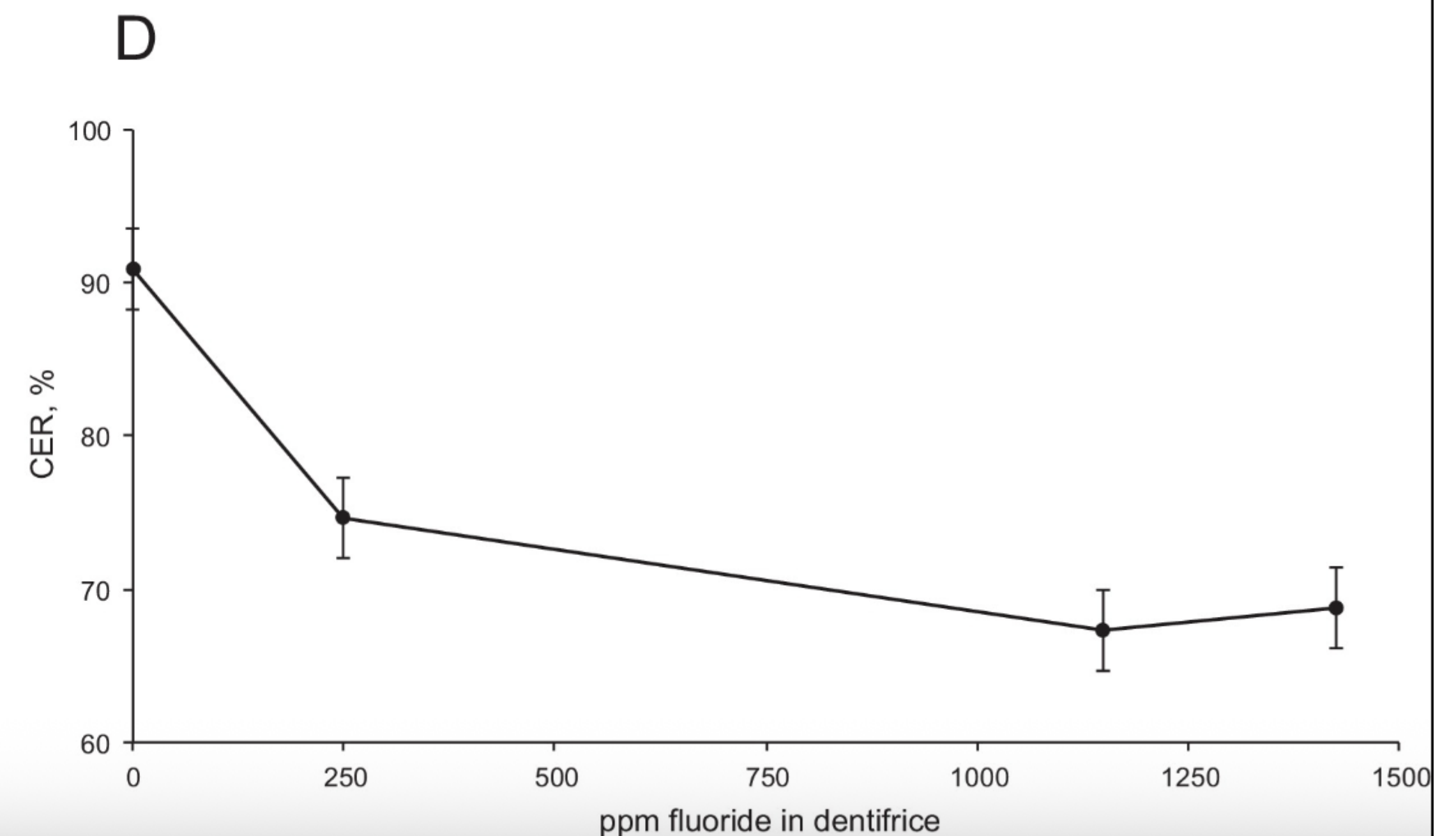
Became less negative: Enamel treated with higher fluoride was significantly more resistant to further erosion



Fluoride Concentration and Mineralization



Significant increase: More fluoride was incorporated into the enamel surface with increasing concentration



Decrease shows higher fluoride concentrations made enamel much more resistant to acid after remineralization





ADA Fluoride Guidelines

Age Group or Dentition Affected	Professionally-Applied Topical Fluoride Agent	Prescription-Strength, Home-Use Topical Fluoride Agent
Younger than 6 years	2.26% fluoride varnish at least every 3 to 6 months ● In Favor	
6-18 years	2.26% fluoride varnish at least every 3 to 6 months ● In Favor OR 1.23% fluoride (APF*) gel for 4 minutes at least every 3 to 6 months ● In Favor	0.09% fluoride mouthrinse at least weekly ● In Favor OR 0.5% fluoride gel or paste twice daily ● Expert Opinion For
Older than 18 Years	2.26% fluoride varnish at least every 3 to 6 months ● Expert Opinion For OR 1.23% fluoride (APF*) gel for 4 minutes at least every 3 to 6 months ● Expert Opinion For	0.09% fluoride mouthrinse at least weekly ● Expert Opinion For OR 0.5% fluoride gel or paste twice daily ● Expert Opinion For
Adult Root Caries	2.26% fluoride varnish at least every 3 to 6 months ● Expert Opinion For OR 1.23% fluoride (APF*) gel for 4 minutes at least every 3 to 6 months ● Expert Opinion For	0.09% fluoride mouthrinse daily ● Expert Opinion For OR 0.5% fluoride gel or paste twice daily ● Expert Opinion For

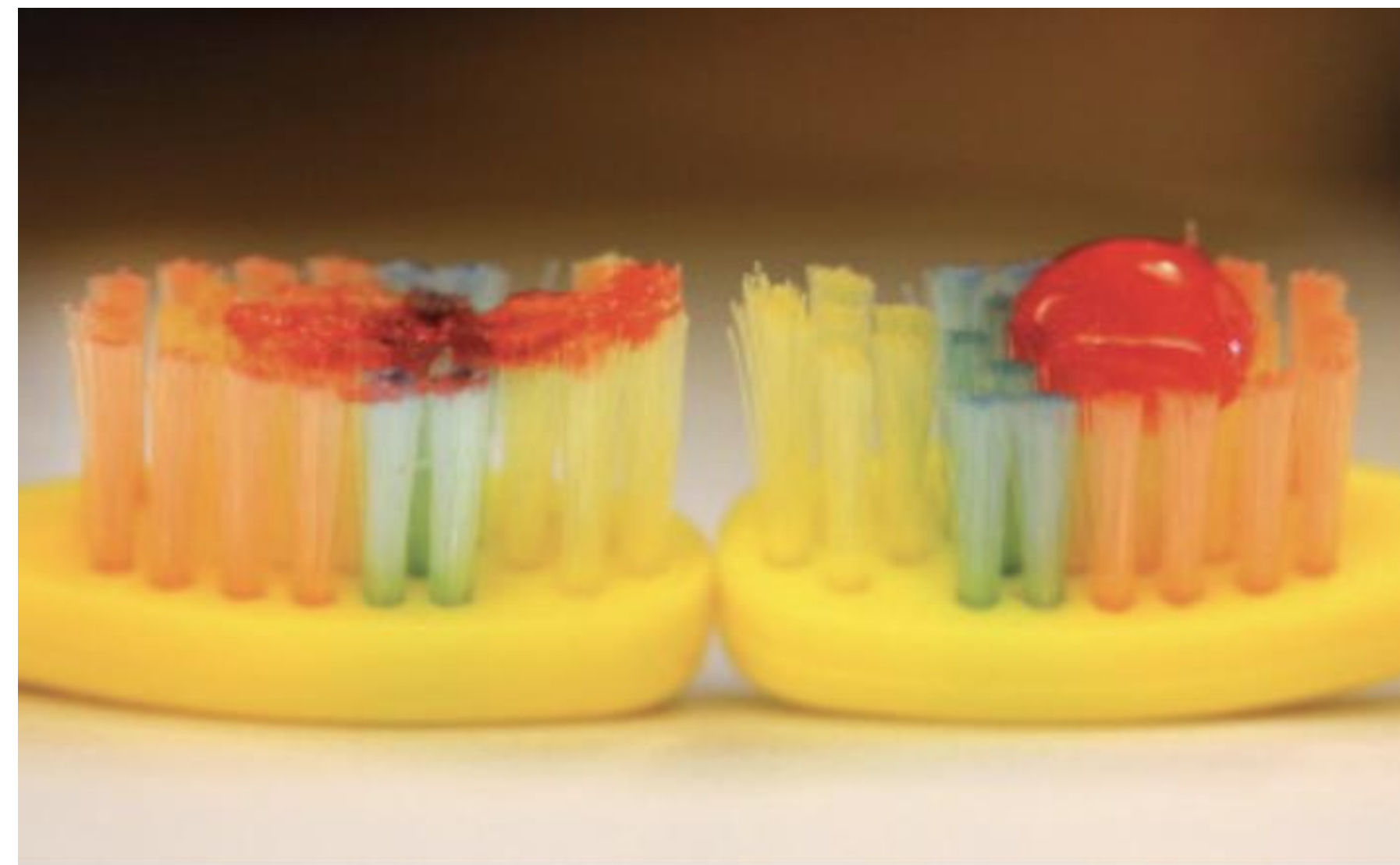
- The ADA recognizes the use of fluoride and community water fluoridation **as safe and effective** in preventing tooth decay for both children and adults.
- The single dose (consumed at one time) of fluoride that could cause acute fluoride toxicity is **5 mg/kg** of body weight (11 mg/kg of body weight of sodium fluoride)



ADA Fluoride Guidelines for Young Children

< 3 yo

"...begin brushing children's teeth **as soon as they begin to come into the mouth by using fluoride toothpaste** in an amount no more than a smear or the size of a grain of **rice**."



3 – 6 yo

"... should dispense no more than a **pea-sized amount of fluoride toothpaste**. Supervise children's brushing to minimize swallowing of toothpaste."



Behavioral Objective 4

Compare and contrast the clinical effectiveness of fluoride delivery systems (varnishes, rinses, pastes, SDF, GI restoratives) based on scientific evidence.

Rachel Dobbs Wineman

Search Strategy



P: Dental Caries

I: Fluoride Delivery System

C: No intervention

O: Clinical Effectiveness

Databases used

Pubmed

Google Scholar

Keywords

Fluoride, Rinse, SDF, Toothpaste

Pubmed search string

((Fluoride) AND (comparison)) AND (SDF) or
(toothpaste) or (Rinse)

Search results

81 results for rinses

210 results for toothpaste

28 results for SDF

Highest level of evidence

Systematic reviews, Clinical
control trials





Fluoride Source	Compared to
SDF	Varnish and Hall Technique
Glass Ionomer	Chitosan loaded GLC, and Fluoride releasing composite
Fluoride Varnish	Pit-and-fissure sealant and resin infiltration
Toothpaste	1 000 ppm NaF vs 1 450 ppm NaF HAF-containing toothpaste
Rinses	Placebo



Silver Diamine Fluoride (SDF)

How does it compare?

- SDF appears to be **more potent** than standard fluoride varnishes for **arresting** dentinal caries (Lowest CFU count; $p=0.01$) (*Alowid et al. 2024*)
- Moderate-certainty evidence shows that SDF **prevents new root caries** compared to no treatment/placebo. (*Worthington et al. 2024*)
- SDF has **lower success** than Hall Technique and SMART, when followed after 12 months. (*Sahebalam et al. 2025*)



BEFORE SDF TREATMENT

AFTER SDF TREATMENT

Source: Google

Fluoride-Loaded Glass Ionomer Restoratives



Source: google

How does it compare?

- Specimens modified GIC with **chitosan and fluoride-loaded nanochitosan** showed increased **Ca, P, and F** in root dentin adjacent to the restoration (*Altınışik et al. 2024*)
- GIC restorations modified with fluoride-loaded chitosan nanoparticles were **more resistant to cariogenic biofilm** challenge than conventional GIC, thus preventing secondary caries formation. (*Altınışik et al. 2024*)
- Fluoride-releasing composite resins offer superior mechanical and esthetic properties, but typically **exhibit significantly lower fluoride** release compared to GICs, RMGIs, or compomers. (*Morawska-Wilk et al. 2025*)

Fluoride Varnish

How does it compare?

- **No statistically significant** difference in efficacy between pit-and-fissure sealants and fluoride varnish in preventing occlusal caries of first permanent molars up to 24 months post-application. (*Han et al. 2025*)
- Both DL (Diode Laser) and the combination of DL and NaF varnish were **more effective** than NaF varnish alone in reducing DH (Dental Hypersensitivity). (*Mohammadian et. al. 2025*)
- Resin infiltration during orthodontic treatment was **superior to topical fluoride** in both arresting the enamel lesions but also significantly improving esthetics of the demineralized regions. (*Kashash et al. 2024*)



Source: google



Source: google

Fluoride Toothpaste

How does it compare?

- All fluoride toothpastes led to significant lesion **depth reduction** over 12 months. (*Wan Hassan et al. 2025*)
- No statistical superiority among **1000 ppm NaF**, **1450 ppm NaF**, or varnish adjuncts. (*Wan Hassan et al. 2025*)
- Regular brushing (2× daily) sufficient to **promote remineralization** (*Wan Hassan et al. 2025*)
- HAF-containing toothpaste may be a **better choice** for children with active caries lesions in primary dentition than F-containing toothpaste. (*Cocco et al. 2025*)



Fluoride Rinses

How does it compare?

- **27% reduction in decayed, missing and filled tooth surfaces in permanent teeth with fluoride mouthrinse** compared with placebo or no mouthrinse. *(Marinho et. al. 2016)*
- Supervised use of a fluoride mouthrinse in children and adolescents is associated with a **large reduction in caries increment tooth decay in the permanent teeth.** *(Cheng, L. L. 2017)*



Behavioral Objective 5

Apply a biologically informed clinical decision tree to **select** fluoride interventions tailored to patient-specific factors such as salivary function, biofilm activity, caries risk profile, and restorative requirements.

Rachel Dobbs Wineman

Search Strategy



PICO not applicable to this BO

Databases used

Pubmed

Google Scholar

Keywords

Fluoride, Rinse, SDF, Toothpaste

Pubmed search string

((Fluoride) AND (comparison)) AND (SDF) or
(toothpaste) or (Rinse)

Search results

81 results for rinses

210 results for toothpaste

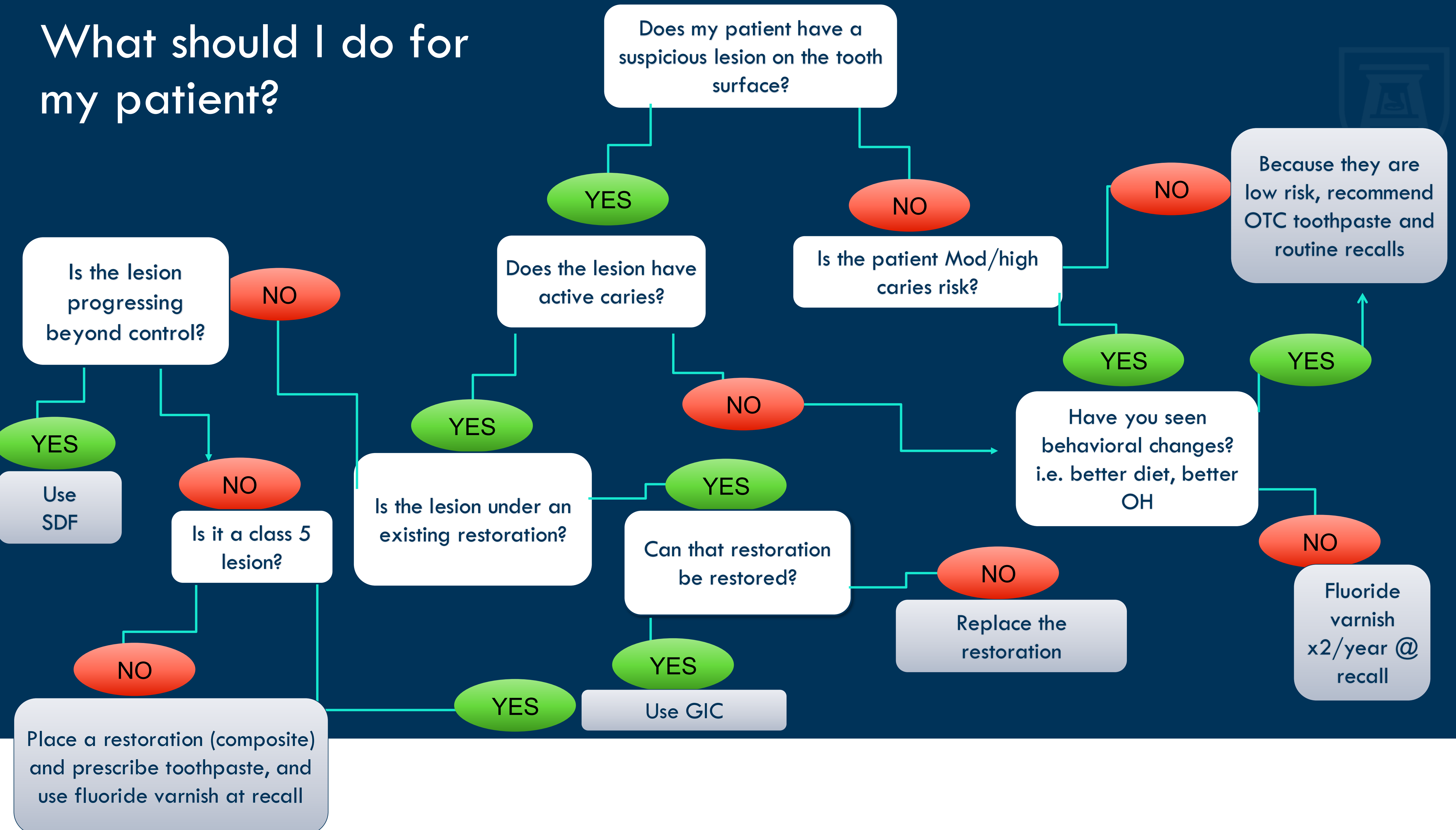
28 results for SDF

Highest level of evidence

Systematic reviews, Clinical control trials



What should I do for my patient?



Fluoride vs. Biofilm: How Everyday Materials Fight Caries

Group 5

Lauren-Ashley Asbridge, Kimberly Clark, Caroline Dudish, Khalil Elhazmiri, Lillian Fantz, Chunghyun Lee, Kellen Norton, Hannah Gresham Putnam, and Rachel Dobbs Wineman

Faculty Mentor: Dr. Jauregui





Case Studies

Chunghyun Lee & Hannah Gresham Putnam





Case Study #1

- An 11-year-old patient presents to clinic for a prophylaxis appointment
- Deep pit and fissures were discovered on the first molar during routine examination
- Patient is in early stages of orthodontic treatment and will need a preventive in-office and at-home care.



<https://dentaladvisor.com/wp-content/uploads/2024/09/CCR43-3M-Clinipro-FINAL.pdf>



Case Study #1

- An 11-year-old patient presents to clinic for a prophylaxis appointment
- Deep pit and fissures were discovered in first molars during routine examination
- Patient is in early stages of orthodontic treatment and will need a preventive in-office and at-home routine.

Is the tooth carious or non-carious?

- A. Carious: Eliminate caries and restore with fluoride-releasing material
- B. Non-carious:
 - Deep Fissure: Fluoride-releasing sealant
 - Sound enamel: Routine care with fluoride at home + Recall



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Comparing resin sealants vs. GI sealants

A. Resin sealant

- If isolation is possible
- Micro-mechanical retention (tighter seal)
- More resistant to occlusal wear

B. GI sealant

- If moisture is present
- Chemical bond
- Less wear resistant

https://www.ejpd.eu/wp-content/pdf/EJPD_2018_19_3_15.pdf



Case Study #1



Tooth was isolated with rubber dam and etched for 15 seconds with 35% phosphoric acid



Sealant was applied into the pits and fissures



Sealed tooth was light cured for 20 seconds.

<https://dentaladvisor.com/wp-content/uploads/2024/09/CCR43-3M-Clinipro-FINAL.pdf>



Case Study #1

Follow-up Care

- 5% Sodium Fluoride Varnish (every 3 months) – release fluoride and reduces bacterial adhesion
- Toothpaste [0.21% NaF + TCP) (2x/day) - release fluoride and promotes tooth remineralization
- Prevent white spots for orthodontic patients



<https://dentaladvisor.com/wp-content/uploads/2024/09/CCR43-3M-Clinipro-FINAL.pdf>



Case Study #2

- Comparative evaluation of the "Remineralizing Efficacy of Fluoride Varnish and Combination Varnishes on White Spot Lesions in Children"
- Evaluated lesions using ICDAS II and its dimensions through photographic method
- Group I (5% NaF, "Profluorid"), Group II [5% NaF with amorphous calcium phosphate (ACP) "Enamel Pro"], and Group III [5% NaF with casein phosphopeptides – amorphous calcium phosphate (CPP –ACP) "MI Varnish"]



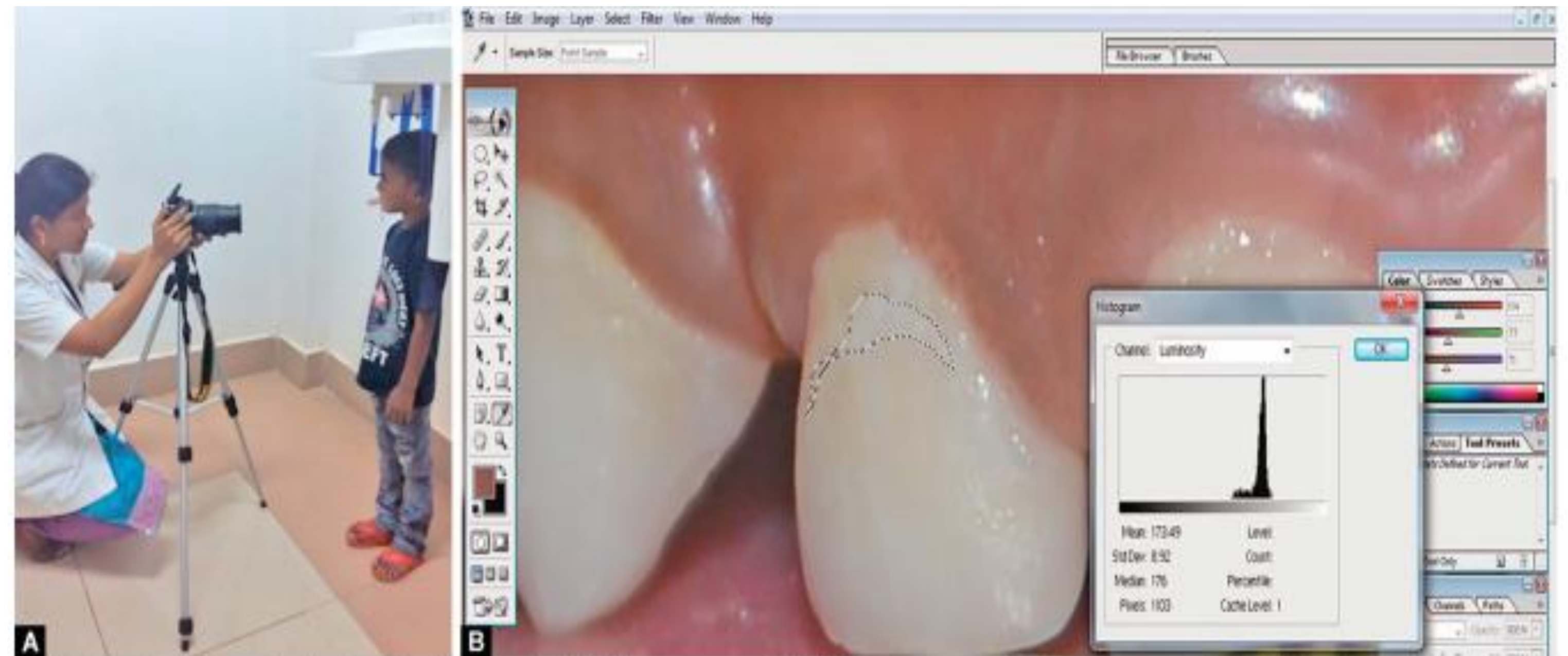
Fig. 1: Assessment of white spot lesion activity—preoperative photograph

<https://www.ijcpd.com/doi/pdf/10.5005/jp-journals-10005-1673>



Case Study #2

- First, oral hygiene instructions and diet counseling were given followed by application of fluoride varnishes in their respective groups.
- The same parameters were recorded at follow-up of 2, 4, 12, and 24 weeks intervals.



Figs 2A and B: (A) Standardization of photographs; (B) Measurement of WSL dimension on displayed images using Adobe Photoshop software

<https://www.ijcpd.com/doi/pdf/10.5005/jp-journals-10005-1673>



Case Study #2



- Overall, the active White Spot Lesions changed to inactive over a period of 24 weeks in GI was 90%, GII was 95%, and 100% in GIII.
- 5% NaF with casein phosphopeptides and amorphous calcium phosphate (CPP –ACP) ["MI Varnish"] was the most effective.
- These combination varnishes could prove to be a promising preventive treatment.



Fig. 3: Postoperative photograph showing resolution of WSL in maxillary primary anterior teeth

<https://www.ijcpd.com/doi/pdf/10.5005/jp-journals-10005-1673>

Behavioral Objective Review

- 1. Describe the biological structure of enamel and dentin and explain how cariogenic biofilms initiate demineralization through acid production, extracellular polysaccharides, and ecological shifts in microbial communities.**

- Enamel is hard and mostly mineral, while dentin is softer, has collagen, and connects to the pulp. When bacteria in plaque eat sugars, they make acid that lowers pH and dissolves enamel minerals. The sticky biofilm traps acid, allowing acid-tolerant bacteria to thrive and continue demineralization.**

Behavioral Objective Review

- 2. Differentiate the microbiological and biochemical mechanisms by which fluoride agents (NaF, SnF₂, CaF₂, SDF, varnishes, pastes, rinses, and GI release) disrupt biofilm metabolism, reduce acidogenicity, and facilitate remineralization.**
- Fluoride slows bacterial acid production and helps rebuild stronger, acid-resistant enamel (fluorapatite). NaF provides fluoride ions, SnF₂ adds antimicrobial effects, and SDF stops decay but stains teeth. CaF₂ layers release fluoride slowly, and varnishes, rinses, pastes, or glass ionomers provide protection at different strengths and durations.**

3. Analyze the dose-response relationship of fluoride by identifying concentrations and forms required to inhibit biofilm activity, enhance mineral uptake, and maintain biological safety across patient populations.

- Fluoride has a dose-dependent effect on both inhibiting cariogenic biofilms and enhancing enamel remineralization. NaF (250–2000 ppm) reduced *S. mutans* growth and acid production, with higher levels needed for mature biofilms. Dentifrices up to 1500 ppm increased enamel mineral uptake and acid resistance. Overall, moderate fluoride levels ($\approx$ 1000–1500 ppm) support remineralization, while higher concentrations may be required to suppress mature biofilms.

Behavioral Objective Review

4. **Compare and contrast the clinical effectiveness of fluoride delivery systems (varnishes, rinses, pastes, SDF, GI restoratives) based on scientific evidence.**
 - **Toothpaste is essential for everyone. Varnish offers strong, lasting protection for high-risk patients, rinses add moderate help, and SDF arrests decay but leaves stains. Glass ionomer fillings protect nearby areas through slow fluoride release. Treatment choice depends on caries risk and patient needs.**

5. Apply a biologically informed clinical decision tree to select fluoride interventions tailored to patient-specific factors such as salivary function, biofilm activity, caries risk profile, and restorative requirements.

- Care depends on caries risk and saliva flow.

 - Low-risk: toothpaste only.

 - Moderate-risk: add varnish or rinse.

 - High-risk or dry-mouth: stronger products like SDF or fluoride-releasing restorations. The goal is effective, safe, and personalized cavity prevention.

Questions?



Six Thinking Hats[®]



White Hat (Faculty): Provides objective data and scientific evidence about the dental material.



Red Hat (Patient): Expresses emotions, concerns, and preferences regarding the material.



Black Hat (Dentist): Evaluates potential risks, challenges, and clinical implications.



Yellow Hat (Sales Rep): Highlights the benefits and advantages of the material.



Green Hat (Manufacturer): Explores innovation, and future possibilities for the material.



Blue Hat (Board of Dentistry): Ensures all perspectives and standards are considered.



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<https://qr.link/v4fUSG>



SCAN ME

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